

Proximity to the Plague: Measuring Locust Exposure and Child Health*

Zemar Hakim[†]

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Abstract

I investigate the early childhood health outcomes of the Kenyan 2020–2021 desert locust invasion. To account for the multi harvest setting, I build a treatment that conditions on the insect’s biological capacity to damage vegetation, the growth stage of the local crop, and subsequent vegetal decline, matching geo-referenced locust observations to the 2022 Kenya Demographic and Health Survey. Children exposed in utero within 30 km of a verified damage event carry weight-for-height z-scores 0.185 standard deviations below the control group, and are 26 percent more likely to be clinically wasted. Effects are strongest when children are exposed in the second and third trimester. Riverine livelihood zones, where households depend heavily on their own harvest and have the least market cushion lose an additional 0.25 standard deviations on weight-for-height beyond the average cropping-zone effect.

Keywords: desert locust, child health, wasting, exposure measurement, Kenya

JEL Codes: I15, O12, Q12, Q54, C18

*All errors are my own.

[†]Department of Economics, Queen’s University.

1 Introduction

Cross-sectional household surveys such as the Demographic and Health Surveys (DHS) and the Multiple Indicator Cluster Surveys (MICS) are widely used to identify the consequences of natural disasters (Michalopoulos and Papaioannou, 2018). As these surveys provide geocoded sample clusters, a researcher can link a household to the shock of interest by drawing a radius around the observed event and treating everyone inside this radial “buffer” as exposed.

A small but growing literature in economics has come to employ these techniques to the incidence of locust plagues (Chatterjee, 2022; Le and Nguyen, 2022; De Vrejer et al., 2015). These papers often assign exposure from a locust’s recorded presence within a radius or an administrative boundary, without conditioning on the insect’s biological capacity to harm vegetation, the phase of a crop’s life cycle or on whether a crop was subsequently damaged following exposure to the plagues. Conte et al. (2023) use a proximity-based measure and rely on Mali’s primarily single harvest to time specific pathways to health from the locusts. However, many countries hit by these types of agricultural shocks are bimodal; their harvests follow from two rainy seasons in a year. The 2020-2021 desert locust invasion mainly hit bimodal East African countries such as Uganda, Ethiopia and Kenya. I estimate its effect on child health in Kenya, a bimodal, multi-crop setting.

I match georeferenced locust observations to the 2022 Kenya Demographic and Health Survey and build an exposure measure that requires the insect to have arrived when a crop was actually standing in the field. Kenya’s bimodal rainfall gives it two planting windows, two harvests, and multiple crop calendars that differ across 24 livelihood zones by several months, so the same calendar date maps to a susceptible crop in one zone and bare ground in another. I use that calendar to gate exposure: a locust record counts only if it falls within a given radius of a survey cluster while the child was in utero, if it describes a gregarious form capable of stripping a field, if the local crop calendar places a crop in a susceptible growth stage on the record’s date, and if satellite vegetation over the cluster then declines relative to that cluster’s own pre-invasion climatology. I build the measure at nine radii from 10 to 50 km.

Children exposed in utero under this measure carry weight-for-height z-scores 0.185 standard deviations below unexposed children in the same survey cluster born in a different month. Adding county-specific linear time trends raises the estimate to 0.207. At the clinical threshold the probability of wasting rises 1.9 percentage points against a baseline of 7.2 percent, a relative increase of 26 percent, and 3.6 percentage points once county trends are absorbed. The estimate is stable across the child, family, and trend specifications and is significant at every specification. It holds at every radius from 10 to 50 km and grows as the radius tightens, reaching 0.313 standard deviations within 10 km.

Wasting records a deficit of weight relative to attained height and responds to adverse nutritional effects in the short term within months. Height-for-age records accumulated growth over the whole of a child’s life and moves slowly. The median child in this sample was 14 months old at measurement, so a disruption confined to gestation and the months around birth has had little time to register in linear growth. Height-for-age z-scores are significant at 30 km when exposed only in the third trimester. The effects on wasting are strongest and significant when children are exposed in the second and third trimester.

The paper also contributes to the fetal origins literature. Barker (1990) proposed that adult disease traces to conditions in-utero, and economists have since documented effects of prenatal shocks on long-term health, schooling, and earnings (Almond and Currie, 2011; Almond et al., 2018). Rainfall and drought supply most of the identifying variation. Maccini and Yang (2009) link early-life rainfall to adult height and schooling in Indonesia, Hoddinott and Kinsey (2001) and Alderman et al. (2006) document persistent growth faltering after Zimbabwe’s droughts, and Shah and Steinberg (2017) show that rainfall raises human capital in utero

while lowering it at school age. Anthropometric z-scores are the standard outcome because early childhood growth predicts adult human capital (Victora et al., 2008). Rainfall shocks identify the effect of income and food availability on the fetus. But they cannot separate the agricultural channel from the waterborne-disease channel, the maternal-labor channel, or the price channel. My exposure measure isolates the agricultural channel by requiring a damaging swarm, a crop in the ground, and a subsequent decline in vegetation.

The remainder of this paper is organized as follows. Section 2 describes the locust, Kenya’s agricultural landscape, and the 2020–2021 invasion. Section 3 describes the data. Section 4 builds the exposure measure and the estimating equation. Section 5 reports results and robustness. Section 5.4 examines mechanisms. Section 6 concludes.

2 Setting

2.1 The Desert Locust

The desert locust, *Schistocerca gregaria*, is a grasshopper that changes phase with population density. At low densities it is solitary (solitarious). These locusts avoid one another, move relatively little, and typically remain within a restricted area, dispersing rather than congregating. Their coloration is usually cryptic, matching the background vegetation and soil, and their bodies are slimmer with proportionally longer legs. Because they neither aggregate nor migrate in large numbers, solitary locusts cause little or no damage to crops and are not considered a plague threat. This phase persists as long as densities remain low, and it is the state in which locust populations spend most of their time.

In the gregarious phase, by contrast, locusts are crowded and highly social. When rainfall concentrates individuals onto scarce green vegetation, crowding triggers a transformation toward this phase over hours to generations (Symmons and Cressman, 2001). Gregarious individuals aggregate, march as hopper bands before they can fly, and form adult swarms that travel with the wind over hundreds of kilometers. They develop striking black-and-yellow (or pink in immature adults) warning coloration, broader bodies, and shorter legs, and they are more active and voracious than their solitary counterparts. Swarms can contain billions of individuals and consume vast quantities of vegetation in a single day, making the gregarious phase the one responsible for the devastating agricultural damage for which desert locusts are notorious. The two phases are endpoints of a continuous, density-dependent continuum, with a transient phase occupying the middle ground.

Only gregarious swarms and hopper bands reach the densities that strip a field. Solitarious individuals occur throughout the locust’s recession area in most years without agronomic consequence.

Damage also depends on timing. Locusts are polyphagous and consume cereals, legumes, and pasture, but the loss a field sustains depends on what stage the crop has reached. Standing biomass close to harvest cannot be replaced within a season. A field at planting can be resown at the cost of seed and labor. Grain already harvested and stored is not at risk, and a field out of season has nothing to lose. A swarm passing over a harvested field is agronomically irrelevant even though it appears in the record as a locust observation.

Upsurges are episodic. Global control efforts suppressed them for much of the late twentieth century (Zhang et al., 2019), and recent work links renewed activity to extreme rainfall events and their climatic drivers (Salih et al., 2020; Liu et al., 2024; Peng et al., 2020).

2.2 Kenya’s Agricultural Landscape

Kenyan agriculture is organized around two rainy seasons. The long rains fall from roughly March to May and the short rains from October to December. Most of the country’s maize, the dominant food staple, is planted with the long rains in the western and Rift Valley highlands and harvested from August onward. Eastern and coastal areas depend more heavily on the short rains and on drought-tolerant crops including sorghum, millet, and cowpeas. The timing of planting, vegetative growth, and harvest therefore differs across the country by several months, and a locust arrival that is catastrophic in one county may be harmless in another on the same date.

County boundaries do not track how households feed themselves. Figure 1 maps the FEWS NET 24 livelihood zones under the household economy approach, each covering an area whose residents share a production system and a set of food-access strategies ([Famine Early Warning Systems Network \(FEWS NET\), 2011](#)).

Cropping zones cover the western and central highlands, the eastern slope of the highlands, and the coastal strip. They include the Central Highlands High Potential Zone (KE19), the Western Medium and High Potential Zones (KE20 and KE21), the Southeastern mixed farming zones (KE16 and KE17), two coastal farming zones (KE12 and KE13) and two marginal mixed farming zones (KE06 and KE22). Households in these zones grow maize, beans, potatoes, and in the highlands tea and coffee, and buy the balance of their food.

Pastoral zones cover the arid north and northeast and a belt of southern rangeland. They include the Northwestern, Northern, Northeastern, Eastern, Southeastern, and Southern Pastoral Zones (KE01, KE05, KE09, KE10, KE11, KE15). Households herd camels, cattle, and small stock, move with the rains, and obtain food from milk, livestock sales, and purchase.

Agropastoral zones sit on the transition between the two and combine rain-fed cropping with herding (KE03, KE07, KE18, KE24). They hold 986 children. Riverine zones are narrow irrigated strips along the Turkwell, the Dawa at Mandera, and the Tana (KE02, KE08, KE14), holding 196 children. Fishing zones cover the shores of Lake Turkana and Lake Victoria (KE04, KE23).

Locusts consume pasture as well as crops ([Le Gall et al., 2019](#)), so pastoral households are not unaffected, but the mechanism and the timing differ from crop loss. A crop calendar governs when a field is vulnerable, and a zone with no crops has no such calendar. The FEWS NET seasonal calendar for a pastoral zone contains no crop row at all, only livestock production, migration, and market access.

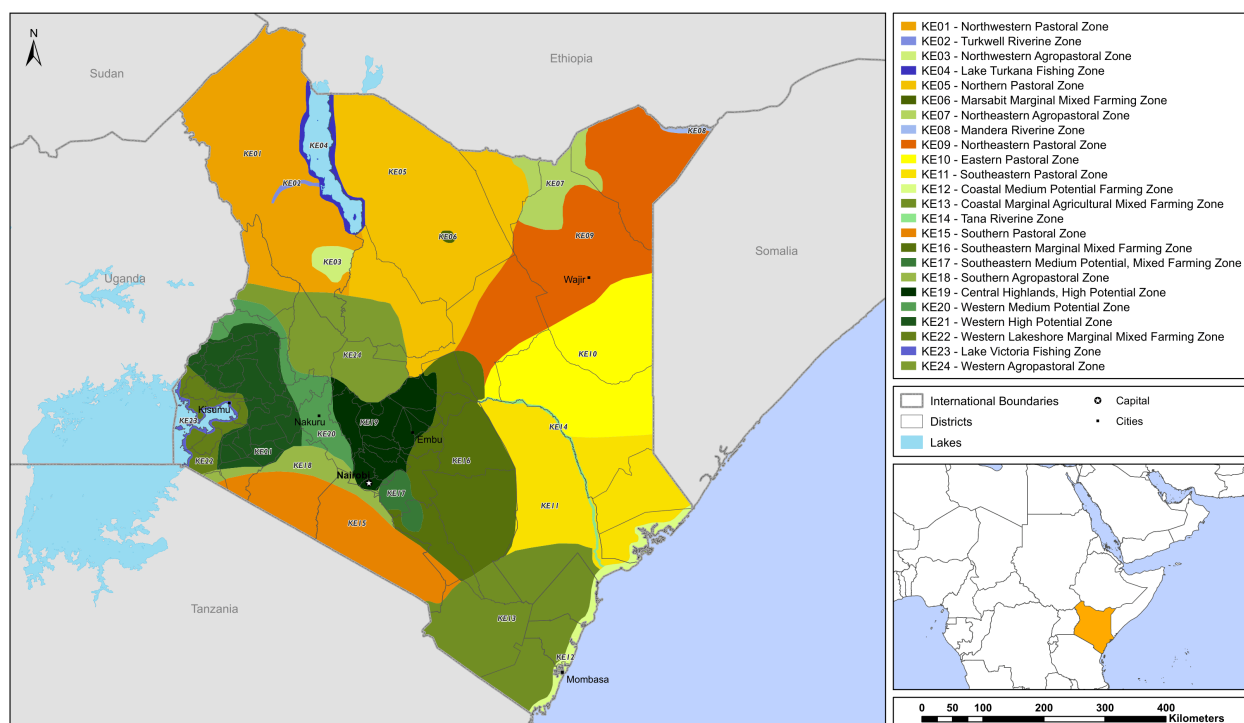


Figure 1: FEWS NET livelihood zones of Kenya. Source: (Famine Early Warning Systems Network (FEWS NET), 2011)

2.3 The 2020–2021 Kenyan Crisis

Heavy and repeated rainfall over the Arabian Peninsula in 2018 and 2019 produced several undetected breeding generations of the desert locust. Tropical cyclones delivered exceptional precipitation to usually arid terrain, creating temporary lakes in the desert and allowing locusts to breed (Salih et al., 2020). These favorable conditions sustained three generations of breeding in roughly nine months, roughly a twentyfold increase over a typical year.

Swarms then crossed into the Horn of Africa through Yemen, Somalia, and Ethiopia, reaching northern Kenya in late December 2019. By early 2020 swarms had spread across much of central and eastern Kenya, prompting the national government to declare an unprecedented threat to food security and livelihoods. The invasion represented the worst locust upsurge Kenya had experienced in seventy years.

The upsurge continued into 2021. A second wave entered Kenya from Ethiopia and Somalia in mid-November 2020 and eventually reached 15 counties, reversing the gains of the preceding months (Ministry of Agriculture, Livestock, Fisheries and Cooperatives, Kenya, 2021). Kenya responded with nine surveillance and spray aircraft, 21 ground spray vehicles, and more than 500 trained personnel, operating alongside FAO-supported control operations (EnviroNews Nigeria, 2021). Ground and aerial spraying proceeded while breeding continued, and control teams worked to suppress new hopper bands before they could fledge into swarms. The campaign brought the invasion largely under control by mid 2021 with some gregarious sightings in the fall of that year. The FAO declared the invasion officially over in February 2022 (Food and Agriculture Organization of the United Nations, 2025).

Figure 2 plots daily counts of gregarious swarm and hopper-band observations across the episode. The invasion arrives in two waves. The first begins in late December 2019, peaks in May and June 2020, and

subsides by August. The second builds from December 2020, peaks in February 2021, and ends in April. The shock therefore ran for sixteen months and crossed two long-rains seasons and one short-rains season.

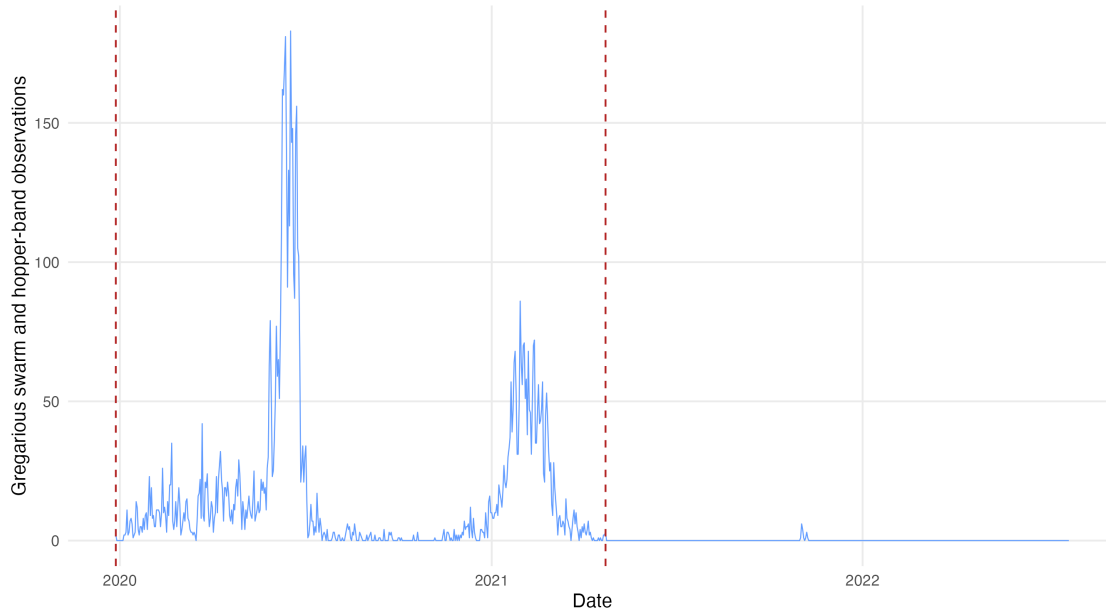


Figure 2: Daily gregarious swarm and hopper-band observations in Kenya. Dashed lines mark the derived invasion window.

3 Data

3.1 Household Survey

The child-level sample is drawn from the 2022 Kenya Demographic and Health Survey (DHS), with the 2014 round used as a pre-invasion falsification sample ([Kenya National Bureau of Statistics \(KNBS\) and ICF, 2023](#); [Kenya National Bureau of Statistics \(KNBS\) and ICF International, 2015](#)). The DHS is a nationally representative household survey conducted under the DHS Program (ICF, formally funded principally by USAID), fielded in Kenya roughly every five years. Each round enumerates all children born to sampled women in the preceding five years, with retrospective birth histories, anthropometric measurement (height and weight, converted to age- and sex-standardized z-scores against the WHO growth reference of [WHO Multicentre Growth Reference Study Group \(2006\)](#)), and household, maternal, and child covariates.

Each DHS cluster (an enumeration area, roughly a village or urban block) is assigned a single GPS coordinate, displaced by up to 2 km for urban clusters and 5 km for rural clusters (up to 10 km for 1 percent of rural clusters, restricted to stay within the same second administrative region) to protect respondent confidentiality ([Burgert et al., 2013](#)). Raster and vector extraction in this paper such as rainfall and distance to roads, is performed as a buffer mean or nearest-feature match over this displacement radius rather than at the point coordinate.

The 7,603 children with valid anthropometry are spread across 1,644 clusters, a mean of 4.6 children per cluster and a median of 4, ranging from a single measured child in the thinnest clusters to 19 in the largest.

3.2 Locust Records

Locust observation records come from the Reconnaissance and Management System of the Environment of *Schistocerca* (RAMSES), the geographic information system used by national Locust Information Officers in each affected country to record survey and control activity, and consolidated by the Food and Agriculture Organization’s Desert Locust Information Service (DLIS) ([Food and Agriculture Organization of the United Nations \(FAO\), 2022](#)). Records are distributed through the FAO Locust Hub. Each record carries a date, coordinates, a presence type (e.g. hopper band, adult swarm, or an explicit absence), a phase classification (gregarious, transiens, or solitary), a maturity classification, and, where control operations occurred, the control type.¹

3.3 Climate and Vegetation

I use three remotely sensed climate products; rainfall and two measures of vegetal health. Rainfall comes from the Climate Hazards group InfraRed Precipitation with Station data (CHIRPS), a quasi-global, 0.05°-resolution, daily-to-monthly precipitation dataset combining infrared satellite imagery with station observations, produced by the Climate Hazards Center at UC Santa Barbara and designed for drought monitoring in data-sparse regions ([Funk et al., 2015](#)). Both monthly and dekadal (10-day) CHIRPS products are used: the monthly product supports gestational rainfall controls and the drought construction described next, and the dekadal product supports growing-season onset detection for the crop calendar.

The Standardized Precipitation Evapotranspiration Index (SPEI), at 6-month accumulation windows, is constructed from CHIRPS precipitation and potential evapotranspiration (PET) from TerraClimate, a global, ~4 km monthly climate and water-balance dataset ([Abatzoglou et al., 2018](#)). SPEI standardises the climatic water balance (precipitation minus PET) rather than precipitation alone, following the index definition in [Vicente-Serrano et al. \(2010\)](#); because the drought conditions overlapping the invasion period combined rainfall deficits with elevated evaporative demand, a precipitation-only index might understate its effects.

Vegetation condition comes from MODIS/Terra Vegetation Indices (MOD13Q1), a 16-day, 250 m composite Normalized Difference Vegetation Index (NDVI) product from NASA’s MODIS sensor, accessed via the AppEEARS extraction service ([Didan, 2015](#)). Pixels are screened on the product’s reliability flag before averaging, so cloud- or snow-contaminated observations never enter a cluster mean. NDVI levels are converted to per-cluster, per-composite z-scores against a pre-invasion climatology.

3.4 Crop Calendars

The exposure measure requires knowing, for a given cluster on a given date, whether a crop was standing in a field and how far it had advanced. Livelihood zone boundaries come from shape files that FEWS NET provides; the FEWS NET Kenya livelihood zone map described in Section 2 ([Famine Early Warning Systems Network \(FEWS NET\), 2011](#)).

The FEWS NET livelihood profiles also published a seasonal calendar for each of Kenya’s 24 livelihood zones, showing land preparation, planting, and harvesting by crop alongside livestock, labor, and market-access cycles ([Famine Early Warning Systems Network \(FEWS NET\), 2011](#)). A cropping zone’s calendar lists maize, beans, potatoes, and the relevant cash crops. A pastoral zone’s calendar lists calving, kidding,

¹Unfortunately, control operation data are sparse and recorded with many errors in this sample.

migration, and milk availability, and contains no crop row. Clusters in zones without a crop row are never placed in a susceptible stage, so they can never satisfy the stage condition.

Each seasonal calendar with a three-color Gantt legend, land preparation, planting, and harvest, plotted month by month for every crop grown in that zone ([Famine Early Warning Systems Network \(FEWS NET\), 2011](#)).² An automated pixel-extraction routine locates the table grid and the legend’s three reference colors on each of the 24 zone pages, then classifies every crop row’s bar by nearest-color match at semi-month resolution. Seventeen zones return at least one crop row. The other seven are auto-detected as having no crop calendar at all, these seven are six pastoral plus the Lake Turkana fishing zone.³ The automated output was checked against an independent manual pixel read of one page, which caught and fixed an off-by-one error in which row’s bar was attributed to which crop, and every zone runs through the identical code path after that fix.

The three-color legend collapses to a single susceptibility ranking. Land preparation leaves bare ground and is not susceptible. Planting is moderately susceptible, since a destroyed planting can be resown at the cost of seed and labor. Harvest, which on these pages denotes the full standing-crop period from establishment through the actual harvest rather than the narrow act of reaping, is the most susceptible stage. A cluster-date outside every crop’s bar on its zone’s page is out of season.

Where a cluster has more than one crop group or more than one season active on a date, the governing stage is the most susceptible of those active, as [Section 4](#) formalizes this.

3.5 Food Prices

Prices come from the World Bank’s Real-Time Prices (RTP) platform, which supplies a monthly price estimate for maize and beans at 233 Kenyan markets from 2007 onward, combining sparse directly-observed transactions with a model that imputes missing values from nearby markets, related commodities, and historical patterns, following the method in [Andrée \(2021\)](#). I take the platform’s own “close” estimate, its best and most current price for a given market-month, as given rather than re-deriving it; I restrict the sample to 2015–2022.

3.6 Roads

Distance from each DHS cluster to the nearest major road is computed from OpenStreetMap’s Kenya road network, exported through the Humanitarian OpenStreetMap Team’s export tool ([Humanitarian OpenStreetMap Team \(HOT\) and OpenStreetMap contributors, 2022](#)). This is a crowd-sourced, continuously updated vector road network.

3.7 Administrative Boundaries

County boundaries come from the geoBoundaries Global Database of Political Administrative Boundaries, distributed via the Humanitarian Data Exchange (HDX) as the Kenya subnational administrative boundaries dataset. Kenya’s 47 counties correspond to the ADM1 level in this dataset, which is produced and maintained by geoBoundaries and published under an open license ([geoBoundaries, 2022](#)).

²Figures [12](#) and [13](#) in [Appendix B](#) provide examples of one calendar for each type.

³Three zones, Tana Riverine, Western High Potential, and Lake Victoria Fishing, needed the corresponding page read by hand instead: the automated color-distance threshold that separates the legend colors was not stable across the environments the pipeline was run in for these three pages specifically, so rather than tune a threshold, those three pages are transcribed directly and flagged as such in the underlying data.

4 Empirical Strategy

4.1 Exposure construction

Let $\mathcal{L}$ denote the set of locust monitoring records, where each record $\ell \in \mathcal{L}$ carries a location, an observation date δ_ℓ , a presence indicator $P_\ell \in \{0, 1\}$, and a damage classification $\kappa_\ell \in \{\text{low, moderate, high, solitary/transiens, none}\}$. Present records in the solitary or transiens phase are classified solitary/transiens regardless of presence type; all other present records are classified low, moderate, or high from their presence type and dominant maturity. A record's *form* is

$$\text{Form}(\ell) = \begin{cases} \text{damaging} & \text{if } P_\ell = 1 \text{ and } \kappa_\ell \in \{\text{low, moderate, high}\} \\ \text{benign} & \text{if } P_\ell = 1 \text{ and } \kappa_\ell = \text{solitary/transiens} \\ \text{absent} & \text{if } P_\ell = 0. \end{cases} \quad (1)$$

Damaging records are present records outside the solitary and transiens phases: swarms, hopper bands, and smaller adult and hopper groups. Benign records are present records in the solitary or transiens phase, whatever their presence type, and are retained throughout as the falsification group in Section 5.3.

For child i born at date b_i to survey cluster $c(i)$, the gestational window is $W_i = [b_i - 266, b_i]$ under an assumed 266-day gestation. For a radius r , the set of records a child could have been exposed to is

$$\mathcal{E}_i(r) = \{\ell \in \mathcal{L} : \text{dist}(c(i), \ell) \leq r, \delta_\ell \in W_i\}, \quad (2)$$

the intersection of geographic proximity and gestational timing. Exposure is indexed by child rather than by cluster, since two children in the same cluster born months apart have different windows W_i and can differ in exposure holding cluster and radius fixed.

4.1.1 Form

The first condition is that the record describe a locust at a density capable of stripping a field. This uses the phase and presence-type classification already coded in the monitoring data through equation (1) and requires no additional assumption. Records that are excluded are carried through the identical construction below as $\text{Form}(\ell) = \text{benign}$ and used as a falsification group.

4.1.2 Crop stage

The invasion struck at different times across the agricultural calendar, and its timing varied from place to place. The May and June 2020 peak fell while the long-rains maize crop stood under its harvest bar in the western zones and was still at planting in the Central Highlands, while the short-rains crop stood under its own harvest bar in parts of the east, and over bare ground under land preparation in counties between seasons. The February 2021 peak fell after the short-rains harvest bar had closed in most of the country. As calendars differ across Kenya by several months, an arrival date on its own carries no information about whether a crop was lost. Two children conceived in the same month, in clusters 200 km apart, can face swarms on the same dates and face different agronomic consequences.

Thus, the second condition is that the arrival find a crop worth destroying. Let $\mathcal{S}(c, \delta)$ be the set of crop stages active in cluster c 's crop calendar on date δ , possibly more than one under staggered planting or

multiple crop groups, and possibly empty. Each stage carries a susceptibility rank,

$$\text{rank}(s) = \begin{cases} 3 & s = \text{harvest (standing crop)} \\ 2 & s = \text{planting} \\ 1 & s \in \{\text{land preparation, out-of-season}\}, \end{cases} \quad (3)$$

and the governing stage on a cluster-date is the highest-ranked active stage,

$$S(c, \delta) = \arg \max_{s \in \mathcal{S}(c, \delta)} \text{rank}(s), \quad S(c, \delta) = \text{out-of-season if } \mathcal{S}(c, \delta) = \emptyset. \quad (4)$$

A stage is susceptible if its rank exceeds 1,

$$\text{Susceptible}(s) = \mathbb{I}[\text{rank}(s) \geq 2]. \quad (5)$$

The harvest stage, which covers the standing crop from establishment through harvest, is susceptible at rank 3. Planting is susceptible at rank 2, since a destroyed planting can be resown at the cost of seed and labor rather than losing the season outright. Land preparation and out-of-season are not susceptible, since a swarm arriving before a crop is planted or after it is harvested has nothing standing to destroy. A cluster in a livelihood zone with no crop calendar has $\mathcal{S}(c, \delta) = \emptyset$ on every date and is never susceptible. Damaging-form arrivals excluded by this step are the same insects in the same places at an agronomically irrelevant time, and are carried through as a second falsification group.

The transcribed FEWS NET calendar fixes a normative planting date for each zone, but the rains that actually trigger planting arrive earlier or later than that norm in any given year and cluster, so the calendar is shifted cluster by cluster to track the rain rather than left pinned to its transcribed date. Season onset is detected separately for each cluster, year, and season from dekadal (10-day) CHIRPS rainfall, searched within a season-specific window, dekads 4 to 18 (roughly early February through late June) for the long rains and dekads 25 to 36 (September through December) for the short rains. Let ρ_d denote a cluster's rainfall total in dekad d of that window. Onset is the first dekad d satisfying three conditions simultaneously:

1. a trigger, $\rho_d \geq 25$ mm;
2. a confirmation, that the one or two dekads immediately following together clear $\sum \rho_{d+1}^{d+2} \geq 20$ mm, so a single wet dekad followed by drought does not count;
3. and a dry-spell check, that at most one of those following dekads itself falls below 5 mm, so a trigger immediately undercut by a dry spell does not count either.

If no dekad in the window satisfies all three, onset is not detected for that cluster-year-season.

A cluster's climatological onset is the median onset dekad across 1981-2018 for that cluster and season. The transcribed calendar boundaries are shifted, holding stage lengths fixed, by the gap between a given year's detected onset and that climatological median, rounded to the nearest dekad and capped at six dekads (60 days) in either direction so a single anomalous year cannot pull the calendar implausibly far from its transcribed position; a cluster-year-season where onset could not be detected is left unshifted at its climatological position rather than dropped. A season that begins three weeks late accordingly shifts that cluster's susceptible window three weeks later rather than leaving it pinned to the calendar date.

4.1.3 Verified damage

The third condition is that vegetation actually suffer. NDVI is observed in 16-day composites. Let $z_{c,t}$ be cluster c 's NDVI anomaly z-score in composite t , and let t_ℓ index the composite containing event ℓ 's observation date. Define the mean anomaly in the two composites preceding and the two composites following the event,

$$\bar{z}_{\text{pre}}(\ell) = \frac{1}{2} \sum_{k=1}^2 z_{c(\ell), t_\ell - k}, \quad \bar{z}_{\text{post}}(\ell) = \frac{1}{2} \sum_{k=1}^2 z_{c(\ell), t_\ell + k}, \quad (6)$$

a window spanning roughly 16 to 48 days after the observation, depending on where within its own composite the event falls. Composites with a missing value are dropped from the corresponding average rather than treated as zero. The change in anomaly is

$$\Delta z(\ell) = \bar{z}_{\text{post}}(\ell) - \bar{z}_{\text{pre}}(\ell), \quad (7)$$

and the event is classified as a verified decline if

$$\text{Decline}(\ell) = \mathbb{1}[\Delta z(\ell) \leq -\theta], \quad \theta = 0.25, \quad (8)$$

fixed in advance. A locust record only counts as "verified damage" if the cluster's greenness fell by at least 0.25 standard deviations (θ).

4.1.4 The exposure indicator

Treatment requires all three conditions to hold on a single record,

$$T_i^{(r)} = \mathbb{1}[\exists \ell \in \mathcal{E}_i(r) : \text{Form}(\ell) = \text{damaging}, \text{Susceptible}(S(c(i), \delta_\ell)) = 1, \text{Decline}(\ell) = 1]. \quad (9)$$

All three conditions must hold on the same observation. A swarm in March and a vegetation decline in September are not evidence that the swarm caused the decline. Equation (9) admits only arrivals followed by a fall in the greenness of the same cluster within roughly six weeks.

Equation (8) uses a variable closer to the outcome than the other two conditions, so three safeguards apply. The test in equation (7) is a change in anomaly rather than a level, so a cluster that was already dry does not qualify on dryness alone. The window in equation (6) is short enough that a later seasonal decline unrelated to the event is not attributed to it. Equations (6) to (8) are applied identically to benign records and to damaging-form records arriving outside the susceptible window. If $\text{Decline}(\ell)$ were capturing drought rather than locust damage, these placebo groups would satisfy it at a similar rate and predict outcomes similarly. Section 5.3 reports that they do not.

Three nested measures are defined by dropping conditions from equation (9). Proximity, $T_i^{(1,r)} = \mathbb{1}[\exists \ell \in \mathcal{E}_i(r) : P_\ell = 1]$, is gated on nothing beyond presence and is the closest analogue to the exposure variables used in existing locust research. Damaging form adds equation (1). Damaging form and stage adds equation (5). Then, by construction $T_i^{(r)} \leq T_i^{(3,r)} \leq T_i^{(2,r)} \leq T_i^{(1,r)}$ and $T_i^{(r)} \subseteq T_i^{(3,r)} \subseteq T_i^{(2,r)} \subseteq T_i^{(1,r)}$.

My measure and the unverified proximity measures treat systematically different children as exposed.⁴

Table 1 compares the disputed children to those both measures agree were exposed. Each difference points to disadvantage among the disputed children. Disputed children have mothers with 3.5 fewer years of

⁴Table 6 reports counts at every radius.

education, live 7.1 km farther from a major road, sit 0.66 of a wealth quintile lower, are 19 percentage points less likely to own land, and experienced gestational conditions 0.44 standard deviations drier on SPEI. Each difference is significant at the 0.1 percent level.

The difference comes mainly from livelihood zones. Among children both measures agree were exposed, 83 percent live in cropping zones, 15 percent in agropastoral zones, and none in pastoral zones. Among disputed children, 35 percent live in cropping zones and 43 percent in pastoral zones. By construction, a pastoral zone has no crop calendar, so no arrival there can satisfy the stage condition, and no child there can be treated. Proximity flags pastoral children at 64 percent at the 30 km radius, higher than the 48 percent rate in cropping zones, because the swarms arrived in the north and northeast first and stayed longest.

Table 1: Characteristics of disputed children, 30 km radius

	Agreed	Disputed	Difference	<i>p</i>
Mother’s education (years)	9.42	5.88	−3.54	< 0.001
Wealth quintile	3.00	2.35	−0.66	< 0.001
Distance to major road (km)	3.75	10.88	7.13	< 0.001
Owens land	0.545	0.359	−0.186	< 0.001
Owens livestock	0.662	0.672	0.010	0.506
Rural	0.640	0.646	0.006	0.715
Gestational SPEI (6 month)	1.265	0.827	−0.439	< 0.001
Gestational NDVI z-score	0.747	0.840	0.092	< 0.001
<i>Livelihood zone shares</i>				
Cropping	0.831	0.352	−0.480	
Pastoral	0.000	0.426	0.426	
Agropastoral	0.145	0.169	0.024	
Riverine	0.020	0.038	0.018	
Fishing	0.003	0.015	0.011	

Agreed denotes children flagged by both the proximity and verified measures ($N = 1,760$). Disputed denotes children flagged by proximity only ($N = 2,310$). *p*-values from two-sample *t*-tests.

Disputed children have a mean weight-for-height of -0.47 , below the -0.09 of children both measures agree were exposed and the -0.10 of unexposed children. The raw means show that the areas proximity adds are drier, poorer, and farther from a road. ⁵

4.2 Health Outcomes

The DHS measures every sampled child’s length or height and weight under a standardized protocol: recumbent length for children under 24 months, standing height above that age, and weight to the nearest 100 grams, up to 59 months inclusive. DHS converts these raw measurements to three z-scores centrally, before the data is released, against the WHO Child Growth Standards, the international reference built from healthy, breastfed children raised without growth constraints ([WHO Multicentre Growth Reference Study Group, 2006](#)). The conversion uses the WHO’s LMS method: at each value of the conditioning variable (age for HAZ and WAZ, height or length for WHZ) the reference population’s distribution is summarized by three sex-specific parameters, a median M , a coefficient of variation S , and a Box-Cox skewness power

⁵Figure 8 shows the counts at every radius.

L , and a child’s raw measurement X is converted to a z-score

$$Z = \begin{cases} \frac{(X/M)^L - 1}{LS} & L \neq 0 \\ \frac{\ln(X/M)}{S} & L = 0, \end{cases} \quad (10)$$

so that $Z = 0$ is the reference median and $Z = -2$ is two standard deviations below it, in the reference distribution’s own units, regardless of the child’s age. I take DHS’s own pre-computed values directly rather than recomputing them, and treat any value outside ± 6 standard deviations as biologically implausible and set it to missing, following the DHS’s own flagging convention.

Weight-for-height (WHZ) compares a child’s weight to the reference for a child of the same sex and the same attained height or length, with age entering only through which reference curve applies, so it nets out the height a child has already reached and asks only whether her current body mass is proportionate to it. This makes it responsive on a short horizon. Since it carries no growth history of its own, WHZ is sensitive to recent nutritional deficits, and the effect of a shock on it decays over the first two years rather than accumulating the way a height deficit does. The clinical label for WHZ below -2 is *wasted*, and a child below -3 is severely wasted.

Height-for-age (HAZ) and weight-for-age (WAZ) are constructed identically, applying the same LMS transformation in equation (10) against the sex-specific WHO reference curves for length-or-height-for-age and weight-for-age respectively, in place of the weight-for-height curve, with the same ± 6 standard deviation plausibility filter. They differ from WHZ, and from each other, only in which anthropometric ratio and which reference curve they use. HAZ compares height to the reference for a child’s age rather than a child’s height, so it is a cumulative record: a child’s height on any given day reflects every month of growth, or growth faltering, since conception, and a single bad month barely moves it. A low HAZ score is a proxy of sustained undernutrition, repeated illness, or a poor environment operating over a long horizon, and the clinical label for HAZ below -2 is *stunted*. WAZ compares weight to the reference for a child’s age; because age enters through both the numerator’s growth history and the denominator’s reference curve, WAZ is a composite of the other two, a child can have a low WAZ score because she is short for her age, because she is thin for her height, or both, and the score alone does not distinguish them. The clinical label for WAZ below -2 is *underweight*.

4.3 Estimation

I estimate

$$y_{ict} = \beta T_i^{(r)} + X'_{ic}\gamma + \alpha_c + \delta_t + \theta_{k(c)} \cdot t + \varepsilon_{ict}, \quad (11)$$

where y_{ict} is the weight-for-height z-score of child i in survey cluster c born in year-month t , $T_i^{(r)}$ is the exposure indicator at radius r defined in Section 4.1, α_c is a survey-cluster fixed effect, δ_t is a birth-cohort fixed effect defined at the year-month level, so that May 2020 and May 2021 occupy separate cells rather than being pooled as the same calendar month, and $\theta_{k(c)} \cdot t$ is a county-specific linear trend in that same year-month index.

The identifying variation is within cluster and across birth cohorts. Two children in the same enumeration area, born eight months apart, face different gestational windows. One window overlaps a verified damage event within r kilometers and the other does not. Equation (11) compares them.

I include survey cluster and birth-cohort (month $\times$ year) fixed effects. The county-specific linear trend

allows counties to sit on different trajectories over the period.

The control set follows [Conte et al. \(2023\)](#). Child controls are sex, birth order, first-born status, twin status, and the interval since the preceding birth. These are the standard determinants of anthropometry that vary within a cluster and within a birth cohort.

Family controls are mother’s education, mother’s height, household head age, household head sex, wealth quintile indicators, and the six-month SPEI over the gestational window. Mother’s height enters because it captures the mother’s own early-life nutrition and is a strong predictor of a child’s growth potential. Wealth quintile indicators enter as a factor rather than a linear term because the relationship with anthropometry is not monotone in the data. Rainfall that is favorable enough to support three generations of locust breeding is also favorable for crops and pasture, so drought conditions are correlated with both the treatment and the outcome. I include SPEI over the child’s own gestational window to remove the part of the estimate that reflects climatic water balance rather than the insects.⁶

Standard errors are clustered on a quarter-degree spatial grid, which yields 370 cells across the sample. These are coarser than the survey cluster. A swarm that arrives at one enumeration area almost always arrives at its neighbors, and the downstream effects on vegetation, local food availability, and prices propagate over a similar area. Residuals are therefore correlated across nearby clusters and not only within them, and clustering at the DHS cluster would overstate precision. The quarter-degree grid gives a wider-than-cluster but still local unit. Clustering at the county instead would leave 47 groups, too few for reliable asymptotic reliability and too coarse to absorb variation.

All regressions are weighted by the DHS individual sampling weight.⁷

5 Results

5.1 Health Outcomes and Radii

Table 2 reports the wasting estimates. Columns (1) to (4) use the weight-for-height z-score. Column (1) includes only the fixed effects, column (2) adds the child controls, column (3) adds the family controls, and column (4) adds county-specific linear time trends. Columns (5) and (6) repeat columns (3) and (4) with an indicator for wasting, a weight-for-height score below -2 , as the outcome.

In utero exposure to a verified damage event within 30 km lowers weight-for-height by 0.207 standard deviations in the most conservative specification. This corresponded to a 3.6 percentage point rise in the probability of wasting in column (6), a 50 percent increase over the baseline of 7.2 percent among unexposed children. Without the county trend the estimate is smaller on both measures, 0.185 standard deviations on weight-for-height and 1.9 percentage points on wasting, a relative increase of 26 percent. The weight-for-height z-score estimate moves by less than 0.02 standard deviations as the child and family controls enter. The standard error is stable near 0.055 throughout and the weight-for-height z-score estimate is significant at the 1 percent level in every column.

Among children verified-exposed at the main radius, the number of confirmed damage events during gestation predicts weight-for-height beyond a simple binary indicator of exposure. Table 3 reports the relationship under three functional forms: the raw count of verified damage events entering linearly, the log

⁶Occupation and food consumption are excluded from the control set. These variables are unfortunately noisy self-reports in the DHS and are largely missing or the timing requested is irrelevant to the study; one week in most cases.

⁷The weight distribution is slightly right-skewed. Appendix B describes it and reports the estimates under winsorized weights (Table 27).

Table 2: In utero locust exposure and child wasting, 30 km radius

	Weight-for-height z-score				Wasted	
	(1)	(2)	(3)	(4)	(5)	(6)
In utero exposure	−0.192*** (0.056)	−0.195*** (0.055)	−0.185*** (0.055)	−0.207*** (0.057)	0.019** (0.008)	0.036*** (0.010)
Child controls	No	Yes	Yes	Yes	Yes	Yes
Family controls	No	No	Yes	Yes	Yes	Yes
County time trend	No	No	No	Yes	No	Yes
Observations	7,443	7,443	7,416	7,416	7,416	7,416
R^2	0.344	0.351	0.352	0.361	0.261	0.270

All regressions include birth cohort and enumeration area fixed effects. Standard errors clustered on a quarter-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent. All regressions weight by the DHS individual sampling weight. Child controls are sex, birth order, first-born status, twin status, and the preceding birth interval. Family controls are mother's education, mother's height, household head age and sex, wealth-quintile indicators, and six-month gestational SPEI. Control coefficients are estimated but not reported. County time trend denotes county-specific linear trends in calendar month of birth. The outcome in columns (5) and (6) is an indicator for a weight-for-height z-score below −2, with a baseline prevalence of 7.2 percent among unexposed children. There are 1,760 treated children.

of one plus that count, and the count capped at 20 to limit the influence of the small number of children exposed to an unusually large number of events. Each additional verified event is associated with a 0.013 standard deviation reduction in weight-for-height, significant at the 5 percent level; the log and capped forms return larger, more precisely estimated reductions of 0.237 and 0.029 standard deviations, both significant at the 1 percent level. All three specifications share the control set and fixed effects of column (3) of Table 2 and additionally control for the number of RAMSES ground visits recorded within 30 km during gestation, which separates the effect of confirmed damage from the effect of monitoring intensity in the same area during the same period.

Table 3: Dose response among verified-exposed children: number of damage events and weight-for-height

Dependent variable: Weight-for-height z-score	Linear	Log	Capped at 20
Verified damage events	−0.013** (0.006)		
Log(1 + verified damage events)		−0.237*** (0.079)	
Verified damage events, capped at 20			−0.029*** (0.010)
Observations	1,459	1,459	1,459

Standard errors clustered on a quarter degree spatial grid in parentheses. ***, ** denote significance at 1 and 5 percent. Sample restricted to children verified-exposed at the main 30 km radius. All columns include the control set and cluster by birth-cohort fixed effects of column (3) of Table 2. I control for the number of RAMSES ground visits during gestation, this counts every recorded visit within 30 km, of any outcome, during the child's gestation window, and is included to separate the effect of confirmed damage from the effect of monitoring intensity in that area during that period.

Table 7 and Table 8 in Appendix A report the same three functional forms for height-for-age and weight-for-age, using the identical verified-exposed sample, control set, and ground-visit control. Weight-for-age moves in the same direction as weight-for-height: an additional verified damage event is associated with a 0.016 standard deviation reduction, significant at the 5 percent level, and the log and capped forms return

reductions of 0.279 and 0.037 standard deviations, both significant at the 1 percent level. Height-for-age point estimates are negative but none is statistically significant in the dose-response functional forms.

I estimate the same specifications on the other two standard z-scores and their clinical thresholds. Height-for-age effects are negative and significant only at 10 km and 25 km radii (5 and 10 percent respectively), so the height result depends on the choice of radius. At 30 km, the point estimate is -0.059 with a standard error of 0.071, and the probability of stunting is unchanged. Weight-for-age falls 0.167 standard deviations, though the probability of being underweight does not change (See Table 9).

Weight-for-height records a deficit of body mass relative to attained height, and it is an indicator of acute malnutrition. Height-for-age records cumulative linear growth from conception onward and responds only slowly, over quarters and years. The invasion ran for sixteen months, from late December 2019 till April 2021, but the median child in the estimation sample was 14 months old at measurement and the mean 14.7 months. For the average child, therefore, the shock occupies the recent portion of a short life (WHO Multicentre Growth Reference Study Group, 2006).

Splitting the sample by age at measurement shows the weight-for-height effect declining with age (Table 10 in Appendix A). At 30 km the estimate is -0.410 for children measured before 12 months, significant at the 10 percent level, -0.200 at 12 to 23 months, significant at the 5 percent level, and 0.122 and insignificant at 24 months or older, where only 174 children are treated. Weight-for-age follows the same pattern. Height-for-age is insignificant before 24 months and -0.609 at 24 months or older, significant only at the 10 percent level. The effect of in utero exposure on weight-for-height is therefore largest in the first year of life and decays over the first two years, which is why it is still visible at a median age of 14 months.

Weight-for-age is a composite of body mass and linear growth against age, so an acute weight deficit registers there but is diluted by the height component that has not moved. The continuous point estimate on weight-for-age is 90 percent of the continuous weight-for-height estimate, and it does not cross the underweight threshold.

Table 11 and Table 12 in Appendix A report the full stepped-control specification for height-for-age and weight-for-age in the same format as Table 2.

I also estimate the three health outcomes at all nine radii (Table 13 in Appendix A). Figure 3 plots the decay with confidence bands. Weight-for-height is significant at every one. The estimate is largest within 10 km, at 0.313 standard deviations, and falls monotonically in the radius till 40 km (40 km at -0.135 and 45 km at -0.141) and then is estimated at 0.119 at 50 km. The number of treated children rises from 728 to 2,285 over the same range. Widening the buffer adds children who were farther from any verified damage event and dilutes the average effect, and the estimate shrinks gradually.

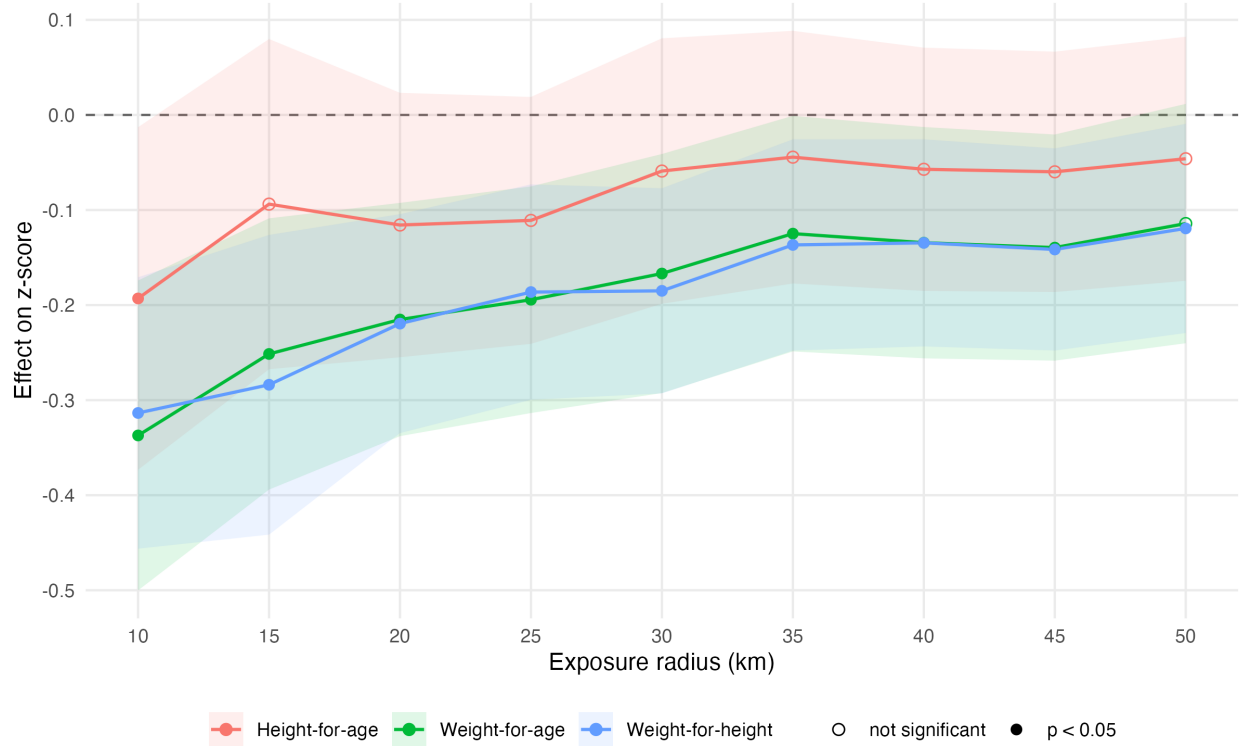


Figure 3: Estimate against radius, by outcome, with 95 percent confidence bands.

5.2 The Exposure Conditions

Figure 4 shows the three conditions of Section 4.1 each regressed at the main radius. Proximity alone has no detectable effect. Its weight-for-height estimate is -0.003 with a standard error of 0.050 , and it flags 4,070 children. Restricting to damaging form removes 79 children and moves the estimate to -0.035 . Adding the crop-stage condition removes 1,263 more and moves it to -0.064 , still insignificant. Adding the verification condition removes a further 968 and moves the estimate to -0.185 , significant at the 1 percent level.

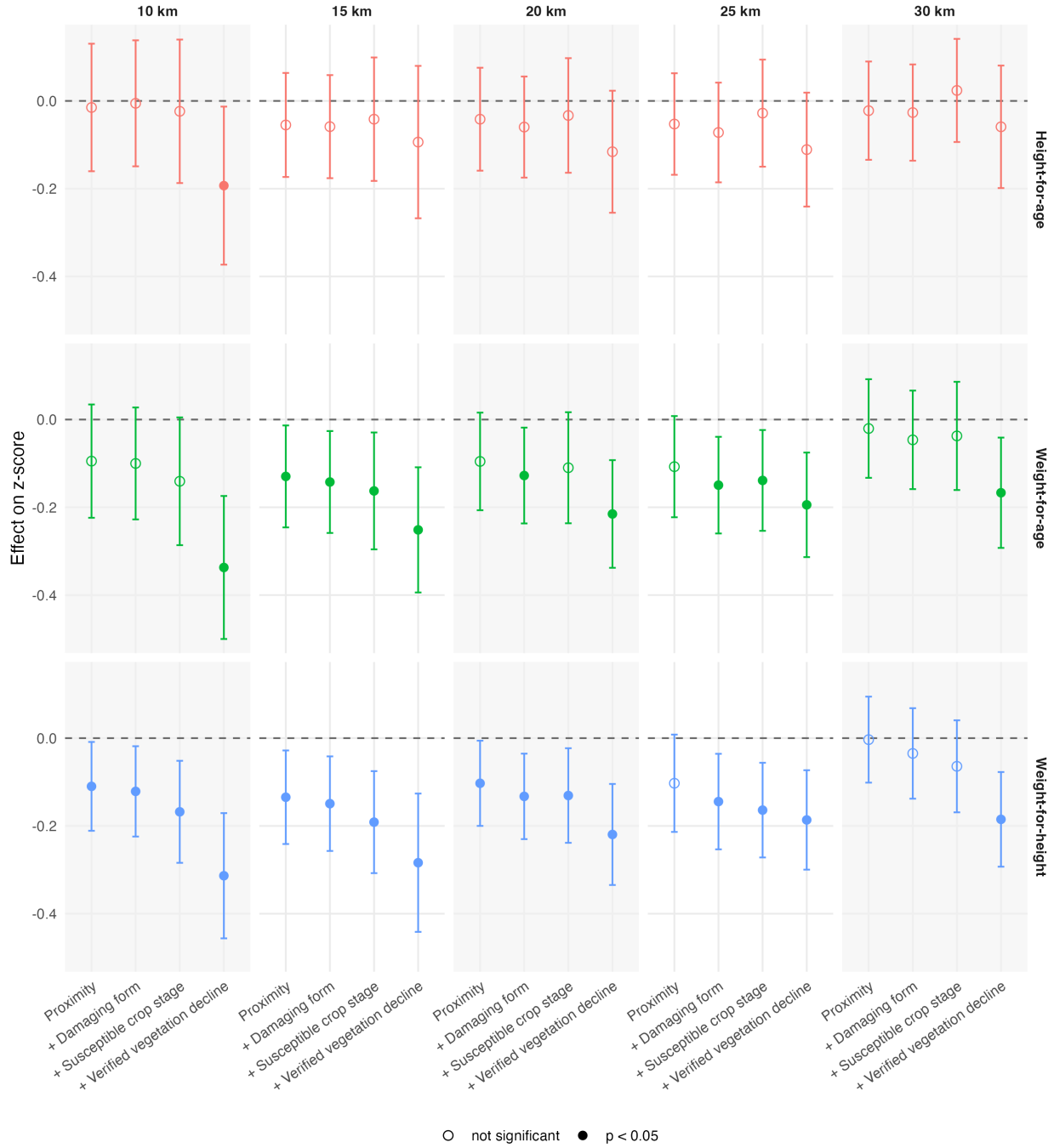


Figure 4: Point estimates with 95 percent confidence intervals.

5.3 Robustness

5.3.1 Falsification Tests

I provide three falsification tests for the exposure conditions. The first replaces damaging-form records with solitary and transiens records and applies the stage and NDVI conditions unchanged. These are locusts at densities incapable of stripping a field. The second uses damaging-form records that arrive outside the

susceptible window, the same insects in the same places when there is nothing standing to destroy.

Applied naively, the solitary placebo flags children who are also flagged by the real treatment, because a cluster visited by solitary locusts during a gestational window can be visited by swarms in the same window. At 30 km, 86 percent of the children the naive placebo flags are also treated. Both placebo groups are therefore purged of any overlap with the treatment before estimation.

Table 14 reports the purged tests. Neither recovers a negative effect on weight-for-height at any radius. The solitary placebo estimate at 30 km is 0.036 with a standard error of 0.098. The out-of-season placebo estimate is 0.085 with a standard error of 0.065. Across 25 purged placebo cells on weight-for-height, none is negative and significant.

Permuting the treatment indicator within survey clusters, 1,000 times, places the 10 km estimate on weight-for-height outside the entire simulated null with a randomization p-value of 0.000. The simulated standard deviation of the permuted estimate is 0.081, close to the clustered analytic standard error of 0.073. Running the same procedure on the solitary placebo returns a randomization p-value of 0.482.

Third, I test whether any loss of greenness during gestation predicts worse outcomes, whether or not locusts were present. I treat every 16-day composite in a child’s own cluster as a candidate event and apply the decline condition of equation (8) and the crop-stage condition of equation (5), without requiring a locust record. The 2014 round gives a direct version of this test, since no vegetation decline in the 2014 cohort’s gestation windows can be locust damage. Declines that meet both conditions are common: 81 percent of 2014 children and 78 percent of 2022 children have at least one during gestation. I therefore also use two other versions: a matched version, which treats the 23 percent of children in each round with the largest decline, the same share as the verified measure at 30 km, and a count of qualifying declines. In 2014, none of the twelve estimates across the three versions and four outcomes is significant at the 10 percent level (Table 15 in Appendix A). The matched estimate on weight-for-height is 0.005 with a standard error of 0.045.

In 2022 the matched decline does predict lower weight-for-height (-0.161 , standard error 0.059), but only where locusts were recorded. Children with a matched decline and a locust record of any form within 50 km during gestation have weight-for-height 0.217 standard deviations lower, while children with a matched decline and no record nearby show no difference (-0.034 , standard error 0.078; Table 16 in Appendix A). Adding the matched decline to the main specification leaves the verified estimate at -0.184 .⁸ A cluster with no locust record is not necessarily a cluster without locusts, so the 2022 comparison is less clean than the 2014 one. In neither round, however, does a loss of greenness away from recorded locust activity predict lower weight-for-height.

5.3.2 Drought

A natural threat to identification is a shock to vegetation that coincides with the invasion rather than following from it. The gestational SPEI control absorbs a smooth, cluster-specific measure of the climatic water balance, but it may not absorb all of it, and if verified exposure is systematically correlated with drought conditions that six-month index leaves unmeasured, the wasting estimate could partly reflect a drought effect misattributed to locusts.

Unconditionally, exposed children come from locations with a somewhat higher six-month SPEI on average than unexposed children, 1.27 versus 0.997 (Table 17 in Appendix A). This reflects where verified locust

⁸In 2022 the count of qualifying declines is also associated with lower height-for-age (-0.038) and weight-for-age (-0.036), both significant at the 1 percent level (Table 15). The count does not separate declines near recorded locust activity from declines away from it, so it does not bear on whether vegetation loss without locusts predicts worse outcomes.

damage happened to occur rather than climatic differences caused by exposure. The gap reverses when the main specification compares children within the same cluster and birth cohort. Exposed children experienced modestly drier gestational windows than their unexposed neighbors in the same cluster and cohort, -0.18 standard deviations on the six-month SPEI index ($p = 0.02$). This within-cluster gap runs in the direction that would make a drought confound plausible in principle. A treated group correlated with drier conditions would bias the exposure coefficient toward the negative result this paper reports even absent any locust effect at all.

I remove the SPEI variable and re-estimate at every radius (see Table 18 in Appendix A). If a residual drought channel not fully captured by SPEI were driving the wasting result, dropping it should let that channel flow unopposed into the treatment coefficient and move it substantially. The estimate changes by 0.005 to 0.008 standard deviations at every radius, and in every case moves marginally larger in magnitude rather than smaller.

Additionally, NDVI and SPEI are not interchangeable measures of the same underlying conditions. SPEI is a climatic water-balance index, precipitation minus potential evapotranspiration, standardized against a location’s own history. It describes moisture stress acting on every hectare in a cluster roughly uniformly, whether or not a crop or pasture stands on it. NDVI records the greenness of the vegetation actually present on the ground, and it responds to anything that removes standing biomass abruptly, not only to a shortfall in the water available to grow it. A locust swarm defoliating a field over a matter of days produces a sharp, localized drop in NDVI on a timescale a slow-moving drought’s drawdown on the water balance does not resemble. Equation (8) is designed to detect a change over roughly six weeks around a specific arrival rather than a seasonal level. This rests on the assumption that no other large-scale consumer of vegetation, a wildlife grazing episode or an unrelated pest outbreak, stripped the same cluster in the same narrow window as a documented locust arrival. Kenya’s rangelands do carry wild herbivores, but a defoliation event that happens to coincide with a verified locust record’s own cluster and six-week window, systematically enough to generate the pattern in Section 5, would have to track the invasion’s own arrival dates and locations rather than occur at random with respect to them.

Equation (8) tests a change in NDVI anomaly over roughly six weeks around a specific locust arrival, not a level of dryness over the season, so a cluster that is dry for the whole gestational window does not qualify on that dryness alone. Both placebo groups sit in the same drought-affected places and the same drought years as the real treatment, so if the decline condition were capturing an ambient dry spell rather than an event-specific loss of vegetation, they should satisfy it and predict the outcome at a similar rate. The test holds the climatic background fixed and asks whether a specific arrival was followed by a decline in vegetation, so it does not depend on the direction of the seasonal SPEI comparison.⁹

5.3.3 Mortality

The main results condition on survival. A child who died before the survey has no anthropometric measurement, so the sample-construction filter that requires a valid z-score removes them. If the shock that depresses weight among survivors also raised mortality risk among the most severely affected, the estimates would understate its severity. If it raised mortality selectively among the children who would have been measured as wasted, the estimates would be biased toward zero.

Additionally, the weaker evidence on an effect on stunting in my analysis might be an artifact of mortality. Stunting is a long-run outcome while wasting is not. Many children might not have survived to an age where

⁹See Figure 9 in Appendix A for more detail.

the long-run measure of health is observed.

I rebuild the birth panel without the anthropometry filter and re-run the identical exposure construction against it. The 2022 round has 8,213 births under this definition. Of these, 7,603 have valid measurements, 263 (3.2 percent) did not survive to interview, and 347 survived but were not measured. The mortality regressions use 8,057 of the 8,213 births: 27 are missing household head age or gestational SPEI, and a further 129 are the only child in their cluster and drop out of the cluster fixed-effects estimation. Table 19 regresses survival on exposure with the same controls, weights, fixed effects, and clustering. Nothing is significant at any radius. Estimates run from -0.001 to 0.005 against a baseline mortality rate of 3.2 percent, and the smallest p-value across the nine radii is 0.48. The three looser measures return the same null at the main radius.

Mortality selection therefore does not explain the weight or height results, and the weight results do not extend to survival.

5.3.4 Pre-existing Conditions

If the exposed clusters were simply on a worse trajectory, a treatment constructed the same way in a pre-invasion period should reproduce the result. I test this using the 2014 DHS round, conducted six years before the invasion.

I shift the locust events back 93 months — the gap between the KDHS 2014 and KDHS 2022 fieldwork periods — landing the invasion’s calendar pattern on the equivalent point in the 2014 cohort’s actual gestation windows, and re-evaluate the stage and NDVI conditions at the shifted dates using rainfall and NDVI from that period. Event locations and 2014 birth timing are unchanged. Whether a 2014 child’s gestational window overlaps the shifted pattern then varies within cluster as real exposure does.

Across four measures, nine radii, and four outcomes — mortality and the three anthropometric measures — 144 cells in total (Table 20 in Appendix A), one is significant at the 5 percent level in the verified measure. This is a mortality estimate that is negative (30km) on an indicator for whether a child died.

5.4 Potential Mechanisms

5.4.1 Anticipation, Crop Failure and Prices

Conte et al. (2023) argue that a locust invasion reaches children through two channels. An anticipatory channel operates during the growing season, when local prices rise on expected crop loss before any loss is realized. A crop failure channel operates afterward, when the harvest actually fails. For Mali they report that children exposed only to the anticipatory channel are harmed about as much as children who also experienced crop failure, and that both channels show up in local crop prices, with an 8.4 percent price increase during the speculative window that fully accounts for the anticipatory health effect once controlled for.

The authors argue that in Mali, the whole country grows crops on one calendar, one planting season, one harvest, once a year. In this setting, when locusts hit in the growing season, there is a shared window where every affected family already knows the harvest is in trouble but the crop has not failed yet.

In Kenya, rain falls twice a year, with many growing seasons for different crops, and the exact timing differs across 24 different farming zones. The invasion also lasted sixteen months across two separate waves. Because no single date can serve as the point of common knowledge, I construct an area-specific boundary date, set by the expected completion of that area’s crop.

For a child to count as "anticipatory," they have to be born strictly after the locust hit (so the family actually knew) but strictly before that same crop's harvest was supposed to finish. Zero children in the sample were born in that narrow slice, at any distance from 10-50 km.

I only count a locust event as confirmed damage if satellite vegetation data shows a drop within 16–48 days afterward. Most verified events occur while the crop is standing in the field rather than at planting: 72 percent of verified event-cluster pairs within 30 km fall in the standing-crop stage. For every exposed child, the expected completion of the affected crop falls before birth, which leaves no children in the anticipatory window.

I attempt to uncover these effects through prices. Price data comes from the World Bank Real Time Prices series for Kenya, 233 markets with monthly maize and bean estimates from 2015 to 2022. The test below uses each reporting market's own price series directly, with no interpolation to a DHS cluster or a county. A market is treated at radius r if a verified event ever fell within r kilometers of it, and its boundary is the harvest completion date of the earliest such event, the same construction used for the health regression above. Calendar time is measured for every market as quarters relative to that boundary rather than as a shared national quarter. A market with no verified event within radius is control throughout its own series. Maize and beans are pooled into one regression since the boundary comes from whichever crop made the governing event susceptible.

$$\log(\text{Price}_{m,t}) = \sum_j \theta_j T_m \times Q_j(t) + \mu_m + \delta_{s,t} + \xi_c \cdot t + \varepsilon_{m,t}, \quad (12)$$

where T_m is the market's own treatment status at radius r , $Q_j(t)$ is the market's own relative quarter, μ_m is a market fixed effect, $\delta_{s,t}$ is a staple by year month fixed effect, and $\xi_c \cdot t$ is a county specific linear trend. Standard errors are clustered on market.

Figure 5 plots $\hat{\theta}_j$ at the main 30 km radius. Two and three quarters before the boundary, treated markets run 2.7 to 3.4 percent above control markets, both significant at the 5 percent level. The coefficient falls toward zero in the quarter before the boundary and the quarter of the boundary itself. From two quarters after the boundary the sign reverses. Treated markets run 3.2 percent below control markets two quarters after and 4.1 percent below three quarters after, both significant at the 1 percent level.

The pattern before the boundary is consistent with an anticipatory price response while the pattern after it is not. A crop failure should raise the price of the crop that failed and the post boundary coefficients intuitively run the wrong way for a supply shock. Locations near a verified event are also more remote on average, and remoteness in this data predicts a thinner transaction coverage in the price series and a systematically lower recorded price. The post boundary result likely reflects that same coverage pattern rather than a market response to realized crop loss. A second reason not to read the post boundary coefficient as a locust price effect is that the 2020 to 2021 exposure window also overlaps the COVID-19 pandemic in Kenya, which brought its own supply disruption and a documented national maize price spike unrelated to any single locust event.

The price data are therefore consistent with the anticipatory channel but not on the crop failure channel. A household in this sample that loses its own crop has less food whether or not market prices change.

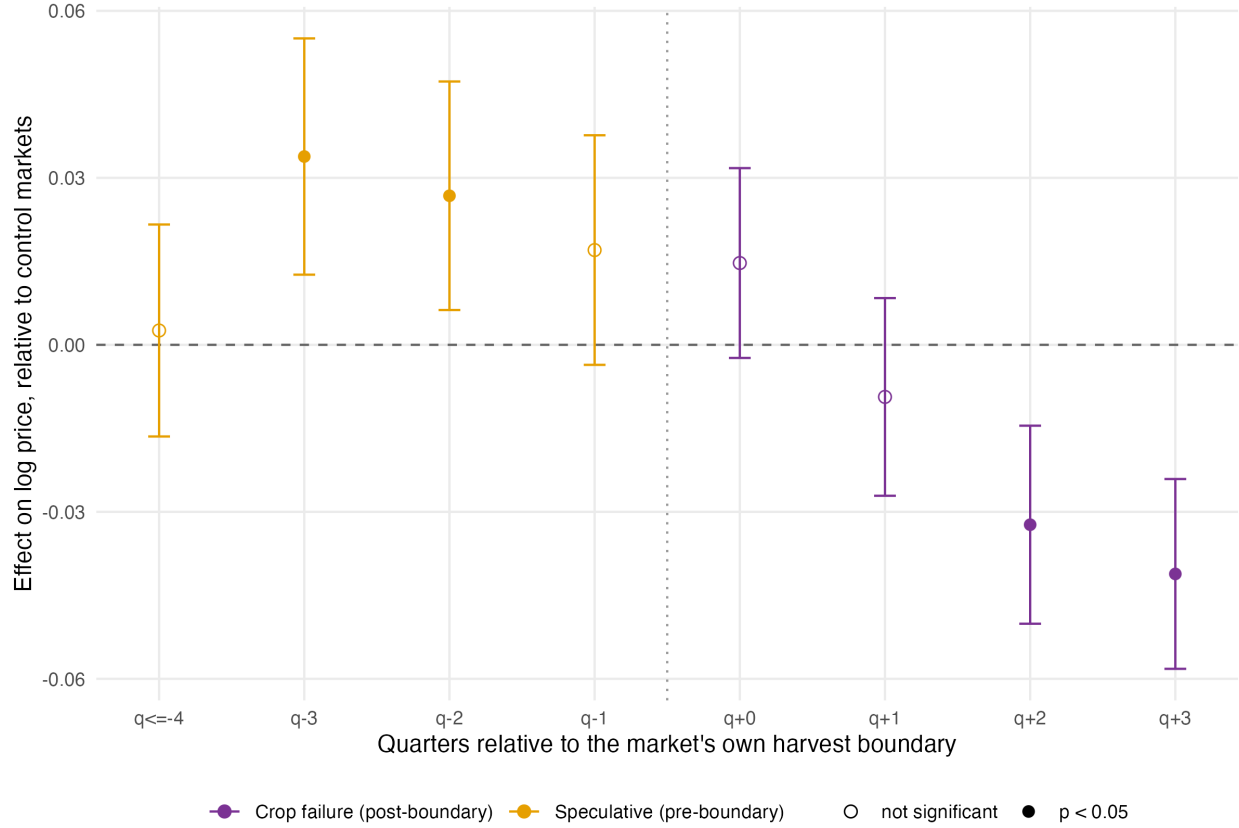


Figure 5: Point estimates with 95 percent confidence intervals. Market level treated versus control price event study. Quarters are relative to each treated market's own harvest boundary.

5.4.2 Psychological Stress of Proximity

Figure 6 shows estimates at restricted samples - pastoral and fishing zones - and regresses each outcome on proximity exposure at every radius. None of the estimates is statistically significant. Point estimates are positive as often as negative and never exceed 0.12 standard deviations, on 642 to 1,203 treated children. These are the households that were closest to the swarms and most often flagged by proximity, yet I find no effect on their anthropometric outcomes.

This is consistent with two things. First, it is consistent with the absence of a general channel operating through the presence of locusts on mental health rather than through physical deprivation from hunger. It is also consistent with the proximity measure's null at the 30 km radius not being a power problem.

I run a similar test on a sample that was only treated when crops are not susceptible. Gregarious swarms of locusts that arrive outside a susceptible crop stage, have no crop-loss mechanism at all, since nothing was standing in the field to threaten. These are "scare" exposures. Run on the cropping zone sample, height-for-age at 50 km is -0.248 ($p = 0.049$) and weight-for-height at 30 km is 0.240 ($p = 0.096$).¹⁰

¹⁰A livelihood zone with no crop calendar can never produce a crop-stage nor verified damage record, so every damaging-form encounter a pastoral child has is biology-only by construction, and pastoral children are already poorer, more remote, and on drier gestational conditions than the rest of the sample.

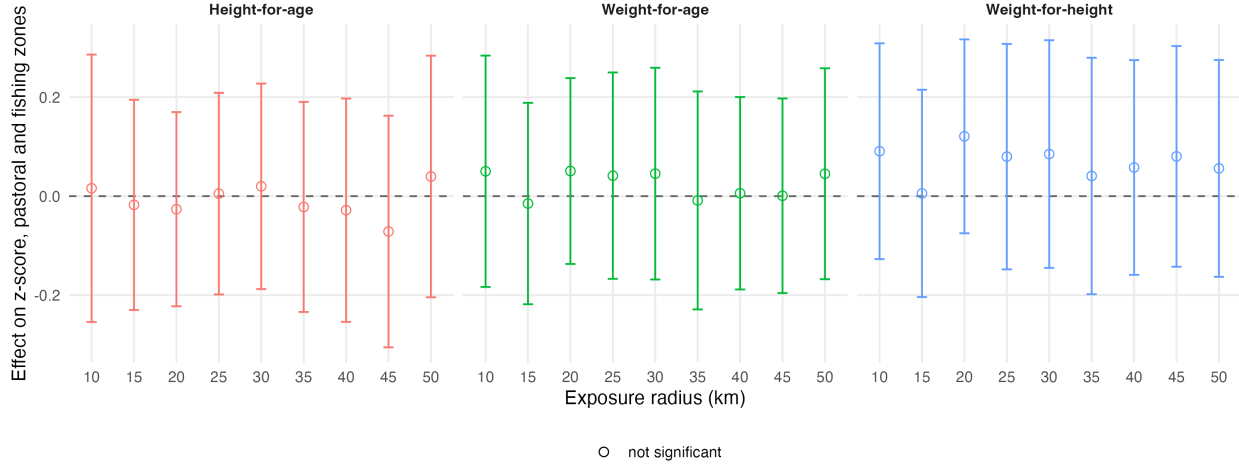


Figure 6: Proximity exposure in pastoral and fishing zones. Point estimates with 95 percent confidence intervals. No estimate is significant at the 10 percent level.

5.5 Heterogeneity

5.5.1 Which Households Are Hit Hardest

Table 4 compares exposed children who are actually wasted to exposed children who are not, at the main 30 km radius. Wasted cases have mothers with 3.0 fewer years of education, sit 0.69 of a wealth quintile lower, and live 1.3 km farther from a major road. Underweight cases show a similar gradient, with mothers 1.2 cm shorter and a 13 percentage point higher rural share (Table 22 in Appendix A). Gestational SPEI is modestly lower among wasted cases (-0.17 standard deviations, $p = 0.118$) and gestational NDVI is modestly higher among underweight cases ($p = 0.015$), but both gaps are small next to the education, wealth, and distance gaps above, which are larger and more consistently significant. At the 15 km radius (Table 23 in Appendix A), neither gestational SPEI nor NDVI differs significantly between wasted and non-wasted children, and among underweight children only the NDVI gap approaches significance ($p = 0.057$).

Table 4: Exposed children who are wasted versus exposed children who are not, 30 km radius

	Wasted	Not wasted	Difference	p
Mother's education (years)	6.56	9.58	-3.02	< 0.001
Wealth quintile	2.34	3.04	-0.69	< 0.001
Distance to major road (km)	4.99	3.69	1.30	0.073
Rural	0.700	0.637	0.063	0.210
Owns land	0.444	0.551	-0.106	0.052
Mother's height (cm)	160.43	160.63	-0.20	0.591
Gestational SPEI (6 month)	1.100	1.274	-0.175	0.118
Gestational NDVI z-score	0.702	0.750	-0.048	0.571
Age at measurement (months)	14.58	15.73	-1.16	0.079

Sample restricted to children flagged as exposed at 30 km. Wasted denotes a weight-for-height z-score below -2 ($N = 90$). Not wasted denotes the remainder of the exposed group ($N = 1,669$). p -values from two-sample t -tests.

Table 5 reports these interactions formally, for all three anthropometric outcomes.

The top panel estimates, separately for each outcome $y \in \{\text{WHZ, HAZ, WAZ}\}$ and each candidate char-

acteristic $Z_i \in \{\text{agricultural occupation, owns land, rural, distance to major road}\}$, twelve regressions of the form

$$y_{ict} = \beta T_i^{(r)} + \lambda Z_i + \eta (T_i^{(r)} \times Z_i) + X'_{ic} \gamma + \alpha_c + \delta_t + \varepsilon_{ict}, \quad (13)$$

one regression per (characteristic, outcome) pair, so the four rows in a single column of the top panel are twelve distinct models, not four coefficients from one shared regression. $T_i^{(r)}$ is exposure at the main radius, and X_{ic} , α_c , and δ_t are the same family-and-child control vector, survey-cluster fixed effect, and birth-cohort fixed effect as column (3) of Table 2; the county-specific trend in equation (11) is not included here. The reported coefficient is η , the interaction between exposure and the characteristic; β and λ are not shown.

The bottom panel instead estimates one regression per outcome that includes every livelihood zone jointly. Let $D_{iz} = \mathbb{1}[\text{zone}_i = z]$ for each zone $z \in \{\text{pastoral, agropastoral, riverine, fishing}\}$, with cropping omitted as the reference zone:

$$y_{ict} = \beta T_i^{(r)} + \sum_z \lambda_z D_{iz} + \sum_z \eta_z (T_i^{(r)} \times D_{iz}) + X'_{ic} \gamma + \alpha_c + \delta_t + \varepsilon_{ict}, \quad (14)$$

with the same control set and fixed effects as equation (13). Cropping's own exposure effect is the coefficient β on $T_i^{(r)}$ alone, not reported in the table; each η_z is that zone's exposure effect relative to cropping's, so the three reported rows in a column all come from the same regression and are directly comparable to one another. No pastoral child is ever treated under the verified-damage measure, so $T_i^{(r)} \times D_{i,\text{pastoral}}$ has no variation to identify η_{pastoral} from and is dropped. Rural residence blunts the exposure effect significantly on weight-for-height and weight-for-age, an offset of 0.24 standard deviations on each, significant at the 1 percent level, though not on height-for-age; agricultural occupation moves the same direction on weight-for-height alone, at the 10 percent level. Land ownership is insignificant and distance to a road shows significance at the 10 percent level for WAZ. These four characteristics are correlated with one another in this sample. Rural residence alone correlates -0.62 with wealth quintile and 0.39 with livestock ownership. They plausibly proxy a single underlying divide between subsistence-oriented and market-dependent households rather than four independent channels. Entering all four jointly in one regression, rather than one at a time (Table 24 in Appendix A), supports this reading. Only rural residence survives, and only at the 10 percent level on weight-for-height and weight-for-age, while agricultural occupation, land ownership, and distance to a road all lose significance once being rural is controlled for directly. Being rural alleviates the treatment effect on short-run weight measures, but the data cannot distinguish whether land, livestock, occupation, or market access accounts for it.

The bottom panel of Table 5 interacts exposure with livelihood zone, with cropping zones as the reference category rather than each zone's own isolated treatment slope, since cropping holds 1,463 of the 1,760 treated children at the main radius and is the natural baseline for the comparison. Riverine zones fare worse than the cropping baseline on all three outcomes; an extra -0.25 standard deviations on weight-for-height ($p = 0.064$), -0.39 on height-for-age ($p = 0.042$), and -0.37 on weight-for-age ($p = 0.007$).

Riverine households are, if anything, less rural than households in the other zones, 31 percent against 60 to 76 percent, so the riverine gap is not a restatement of the rural result. It more plausibly reflects riverine households' dependence on flood-recession agriculture. Riverine households in the underlying livelihood-zone data source roughly 55 percent of their food from their own production, against about 45 percent (Famine Early Warning Systems Network (FEWS NET), 2011) in cropping zones because they have far less of a market cushion when a harvest is lost. Re-estimated at every radius from 10 to 50 km (Table 25 in Appendix A), the riverine coefficient is negative and significant on all three outcomes at 30 and 35 km,

on weight-for-height from 20 km, and on weight-for-age from 15 km, and it is indistinguishable from zero beyond 35 km.

Agropastoral zones show a milder but statistically significant protective effect relative to cropping on all three outcomes, resting on 256 treated children across 80 clusters and 11 counties, though concentrated in the Western Agropastoral zone. Fishing zones return a large, significant positive coefficient on all three outcomes, but it rests on six treated children in a single grid cell in one fishing zone on Lake Victoria.

Table 5: Heterogeneity: 30km exposure interacted with baseline characteristics and livelihood zone

	Weight-for-height z-score	Height-for-age z-score	Weight-for-age z-score
<i>Panel A: Baseline characteristic</i>			
Agricultural occupation	0.212* (0.113)	0.046 (0.134)	0.167 (0.127)
Owns land	0.095 (0.081)	0.006 (0.083)	0.067 (0.082)
Rural	0.239*** (0.088)	0.112 (0.114)	0.239*** (0.088)
Distance to major road (km)	0.011 (0.010)	0.018 (0.011)	0.018* (0.010)
<i>Panel B: Livelihood zone (relative to cropping)</i>			
Agropastoral	0.254* (0.135)	0.329** (0.146)	0.339*** (0.117)
Riverine	-0.246* (0.132)	-0.390** (0.191)	-0.369*** (0.135)
Fishing	0.266*** (0.097)	0.609*** (0.101)	0.475*** (0.114)

Standard errors clustered on a quarter-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent. Panel A: each cell reports the coefficient on exposure (30km) interacted with the row characteristic from a separate regression of the column outcome on exposure, the row characteristic, their interaction, and the control set and fixed effects of column (3) of Table 2. Panel B: each cell reports exposure interacted with livelihood zone from a single regression per outcome, cropping as the reference level, with the same control set and fixed effects; each coefficient is that zone's exposure effect relative to cropping. Pastoral zones are excluded because no pastoral child is ever treated under the verified-damage measure. At the main radius, 1,463 of the 1,760 treated children are in cropping zones, 256 in agropastoral, 35 in riverine, and 6 in fishing; the riverine and fishing coefficients rest on far fewer treated children than the cropping and agropastoral estimates.

5.5.2 Timing Within Gestation

Figure 7 splits verified exposure by the trimester in which the qualifying event falls, at every radius, on all three health outcomes.

First-trimester exposure is never significant on any outcome at any radius. The third-trimester estimate is largest close to the damage and falls with distance. On weight-for-height it is -0.368 at 15 km and -0.174 at 30 km, and from 35 km onward it lies between -0.050 and -0.078 and is insignificant. On weight-for-age it falls from -0.408 at 15 km to -0.211 at 30 km, and beyond 30 km it is at most significant at the 10 percent level (-0.110 at 40 km). The second-trimester estimate does not fall with distance. On weight-for-height it lies between -0.209 and -0.299 and is significant at the 5 percent level or better at every radius from 15 to 50 km, and at the 1 percent level from 25 km. On weight-for-age it lies between -0.173 and -0.259 and is significant at the 5 percent level or better over the same range. At 10 km, where each trimester holds

roughly 250 treated children, neither second-trimester estimate is significant at the 5 percent level.

Which trimester carries the larger estimate therefore depends on the radius. The third trimester is larger at 15 and 20 km on weight-for-height and from 10 to 25 km on weight-for-age. The second trimester is larger at 10 km and from 25 km on weight-for-height, and from 30 km on weight-for-age. At 30 km the second-trimester estimate is -0.299 against -0.174 for the third trimester on weight-for-height, and -0.259 against -0.211 on weight-for-age. The pooled estimate in Table 13 averages over both.

The third-trimester component follows the same distance gradient as the pooled estimate in Section 5. It is large where verified damage was close to the household and shrinks as a wider radius adds children whose households were farther from it. A loss of the household’s own crop during the months of fastest fetal weight gain would produce this pattern.

The second-trimester component does not depend on distance in this way. A child exposed in the second trimester is born roughly three to six months after the event. For every exposed child, the affected crop’s expected harvest was completed before birth (Section 5.4.1), so these children are born after the lost harvest would have been brought in, into a local food shortfall that is not confined to the households whose own fields were hit. A shortfall of this kind reaches beyond the immediate area of a swarm, which would explain why the second-trimester estimate on weight-for-height is about as large at 50 km (-0.209) as at 15 km (-0.234). This interpretation has two limits. The market-level price event study in Section 5.4.1 shows no price increase after the harvest boundary, so if the shortfall operated it would have to run through household stocks, local trade, or transfers the price data do not capture. Next, children born into a shortfall are also exposed to it in early infancy, so the second-trimester estimate may partly reflect postnatal rather than prenatal exposure.

Height-for-age follows the third-trimester pattern. The pooled estimate in Table 13 shows no significant effect on height-for-age at the main radius. Split by trimester, third-trimester exposure is significant on height-for-age from 10 to 30 km (-0.284 to -0.164 , all at the 5 percent level or better), while first- and second-trimester exposure are never significant at any radius. The pooled null at 30 km is an average of a radius-limited third-trimester effect and two null trimesters. The third trimester is the period of fastest fetal linear growth, and Stein and Susser (1975) find a smaller but real effect of third-trimester famine exposure on length at birth alongside the larger effect on birth weight.

5.5.3 Postnatal Exposure

The exposure measure is defined on the gestation window only. A child born in January 2020, for example, lived through the invasion as an infant but is included in the comparison group. If postnatal exposure also lowered weight-for-height, this would bias the estimate toward zero. I define postnatal exposure as a verified damage event within 30 km of the child’s cluster in the first 12 months after birth and before the interview. 1,128 children meet this definition, and 772 of them are not exposed in utero. Adding it to the main specification moves the in utero estimate from -0.185 to -0.145 (standard error 0.064), still significant at the 5 percent level, while postnatal exposure itself is 0.106 (standard error 0.089) and insignificant (Table 21 in Appendix A). Splitting postnatal exposure into the first and second half of the first year gives the same result, and among children not exposed in utero, postnatal exposure is 0.138 (standard error 0.102). Postnatal exposure in the comparison group does not drive the main result.

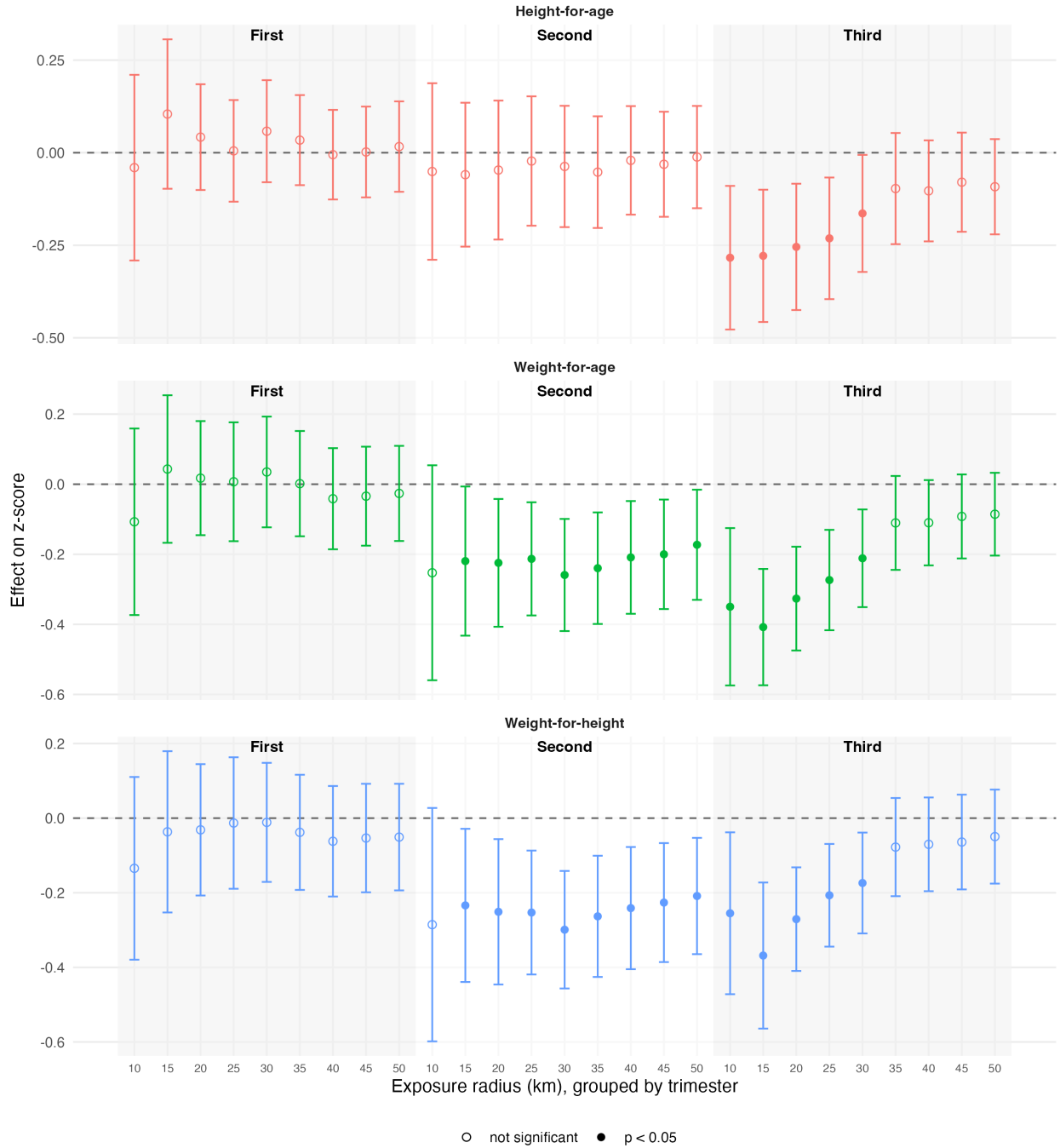


Figure 7: Regressions on an indicator for verified exposure occurring during that trimester, sharing the control set and fixed effects of column (3) of Table 2. Standard errors are clustered on a quarter-degree spatial grid.

6 Discussion

Within 30 km of a verified damage event, in utero exposure lowered weight-for-height by 0.19 to 0.21 standard deviations and raised the probability of wasting by 1.9 to 3.6 percentage points against a baseline

of 7.2 percent. The estimate holds at every radius from 10 to 50 km and is larger at smaller radii. It is concentrated in the second and third trimesters. The third-trimester effect is largest close to the damage and fades with distance, while the second-trimester effect is roughly constant across radii.

These results depend on how exposure is defined. A measure that records locust presence within 30 km of a household returns a coefficient of -0.003 on the same children, outcome, controls, and fixed effects. Requiring the record to describe a gregarious form and to arrive during a susceptible crop stage moves the estimate to -0.064 , still indistinguishable from zero. Requiring a decline in vegetation over the following 16 to 48 days moves it to -0.185 , significant at the 1 percent level. Just over half of the children the proximity measure flags at 30 km show no evidence of vegetation damage around them, and 43 percent of those live in pastoral zones where no crop calendar applies. A swarm that passes over bare ground or rangeland does not destroy a crop, and a measure that counts it as exposure dilutes the estimate toward zero.

The main limitation of this paper concerns the price channel. Section 5.4.1 shows that no children in the sample were born in the anticipatory window I construct, the Kenyan counterpart of the window [Conte et al. \(2023\)](#) use in Mali. The post-boundary price movement in the market-level event study more plausibly reflects thinner transaction coverage in remote treated markets and COVID-era price disruptions than a supply response to crop failure. I therefore cannot separate a price-mediated channel, anticipatory or realized, from the direct loss of a household's own harvest. The health estimates capture whatever combination of the two operated.

The results have implications for policy. A relief program that defines exposure by proximity might enroll 4,070 children where a damage-verified rule enrolls 1,760, and the 2,310 children it adds are from drier, poorer, and more remote households that hold less land and are disproportionately pastoral. The two rules therefore select different populations. A household that lost a harvest needs aid sized to replace that harvest. A poor, remote pastoral household with no crop to lose needs income support rather than compensation for a crop loss. A targeting rule based only on whether a swarm was nearby cannot distinguish the two, and it would direct harvest-replacement aid to households that had no harvest to replace while missing the different needs of pastoral communities.

For research, the results suggest that exposure measures for natural disasters should reflect how the hazard causes damage. These crises damage crops, assets, or health through a specific physical or biological process, and a measure built only on where the hazard was recorded does not capture that process.

Two extensions would add to these results. In the short run, more frequent surveys during a disaster could record how households respond. Knowing whether they sell livestock, change their diets or migrate, or keep harvests they would otherwise have sold would inform policy. In the longer run, following these cohorts could show whether the deficits measured here persist into cognitive development, schooling, and earnings.

A Supplemental Tables and Figures

Table 6: Children flagged as exposed, by measure and radius

Radius (km)	Proximity	+ form	+ stage	+ verified	Disputed	Jaccard
10	2,249	2,140	1,320	728	1,521	0.32
15	2,889	2,792	1,798	1,067	1,822	0.37
20	3,349	3,263	2,148	1,333	2,016	0.40
25	3,721	3,618	2,431	1,529	2,192	0.41
30	4,070	3,991	2,728	1,760	2,310	0.43
35	4,350	4,256	2,962	1,911	2,439	0.44
40	4,628	4,526	3,155	2,046	2,582	0.44
45	4,833	4,727	3,351	2,164	2,669	0.45
50	5,031	4,904	3,503	2,285	2,746	0.45

Sample is 7,603 children. Measures nest, so no child is flagged by a stricter measure and not by a looser one. Disputed denotes children flagged by proximity and not by the verified measure. Jaccard is the overlap between the proximity and verified lists.

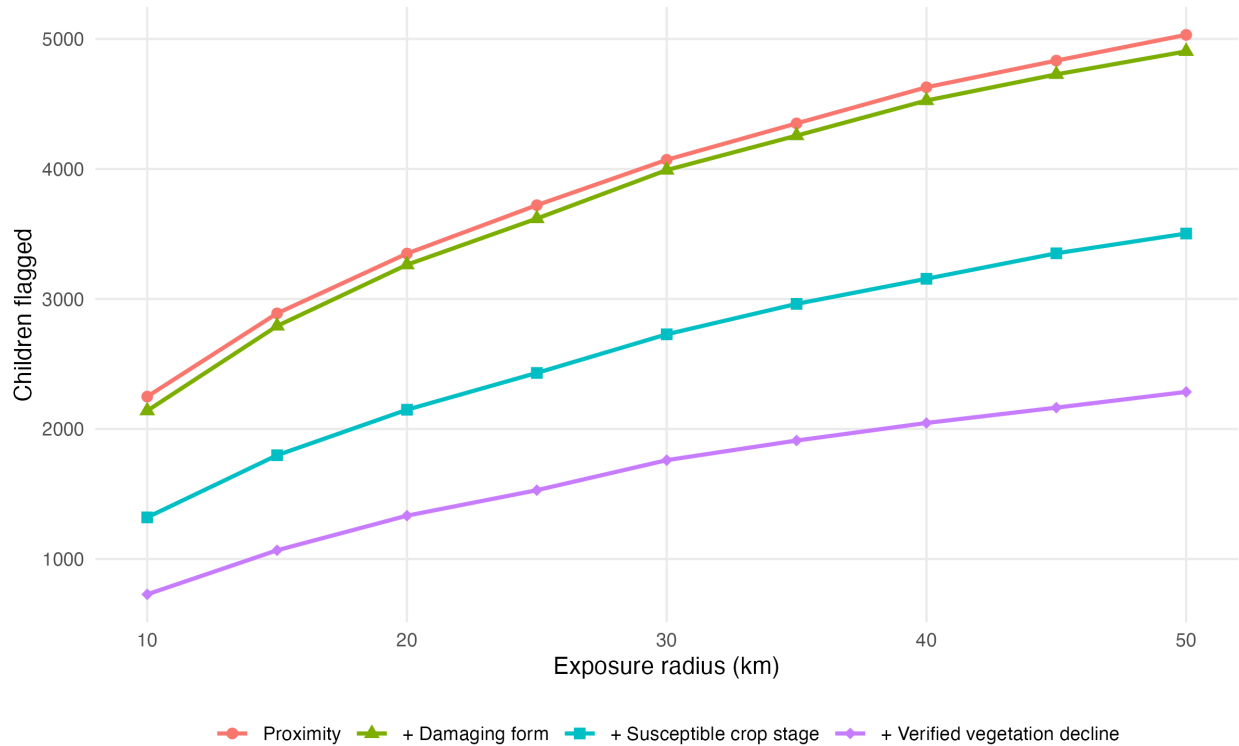


Figure 8: Children flagged as exposed, by exposure definition and radius. The four definitions nest, so every child flagged by a stricter measure is also flagged by every looser measure above it. Reproduces the counts of Table 6.

Table 17 compares exposed and unexposed children on the covariates available before treatment. The unconditional comparison looks like a typical cross-section of Kenyan households sorted by geography: exposed children have better-educated mothers, more land, higher wealth, and live closer to a road, and thirteen of the nineteen variables differ at the 5 percent level. Almost none of this survives conditioning on the same

Table 7: Dose response among verified-exposed children: number of damage events and height-for-age

Dependent variable: Height-for-age z-score	Linear	Log	Capped at 20
Verified damage events	−0.011 (0.009)		
Log(1 + verified damage events)		−0.181 (0.122)	
Verified damage events, capped at 20			−0.025 (0.016)
Observations	1,460	1,460	1,460

None of the three functional forms is significant at conventional levels. Standard errors clustered on a quarter degree spatial grid in parentheses. Sample restricted to children verified-exposed at the main 30 km radius. All columns include the control set and cluster by birth-cohort fixed effects of column (3) of Table 11. Controls for the number of RAMSES ground visits during gestation, counting every recorded visit within 30 km, of any outcome, during the child’s gestation window, to separate the effect of confirmed damage from the effect of monitoring intensity in that area during that period.

Table 8: Dose response among verified-exposed children: number of damage events and weight-for-age

Dependent variable: Weight-for-age z-score	Linear	Log	Capped at 20
Verified damage events	−0.016** (0.008)		
Log(1 + verified damage events)		−0.279*** (0.089)	
Verified damage events, capped at 20			−0.037*** (0.012)
Observations	1,459	1,459	1,459

Standard errors clustered on a quarter degree spatial grid in parentheses. Sample restricted to children verified-exposed at the main 30 km radius. All columns include the control set and cluster by birth-cohort fixed effects of column (3) of Table 12. Controls for the number of RAMSES ground visits during gestation, counting every recorded visit within 30 km, of any outcome, during the child’s gestation window, to separate the effect of confirmed damage from the effect of monitoring intensity in that area during that period.

Table 9: Alternative Anthropometric Outcomes, 30 km radius

	Height-for-age z-score	Weight-for-age z-score	Weight-for-height z-score	Stunted	Underweight	Wasted
In utero exposure	−0.059 (0.071)	−0.167*** (0.064)	−0.185*** (0.055)	0.006 (0.024)	−0.002 (0.013)	0.019** (0.008)
Baseline prevalence				0.185	0.122	0.072
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Cluster and birth-month FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	7,424	7,421	7,416	7,424	7,421	7,416
R^2	0.390	0.377	0.352	0.319	0.275	0.261

Standard errors clustered on a quarter-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent. Each column is a separate regression sharing the control set and fixed effects of column (3) of Table 2. Baseline prevalence is the share of unexposed children below the relevant threshold.

Table 10: In utero exposure by age at measurement, 30 km radius

Age at measurement	Weight-for-height z-score	Height-for-age z-score	Weight-for-age z-score	Treated
0 to 11 months	−0.410* (0.230)	−0.133 (0.222)	−0.400* (0.225)	551
12 to 23 months	−0.200** (0.089)	−0.057 (0.121)	−0.170** (0.085)	1,035
24 months or older	0.122 (0.418)	−0.609* (0.335)	−0.225 (0.328)	174

Each cell is a separate regression on children in the row's age group, sharing the control set and fixed effects of column (3) of Table 2. Observations are 2,371, 3,007, and 711 for weight-for-height, 2,376, 3,012, and 712 for height-for-age, and 2,375, 3,009, and 712 for weight-for-age, by age group. Treated counts the children in each age group flagged by the verified measure at 30 km. Standard errors clustered on a quarter-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent.

Table 11: In utero locust exposure and child stunting, 30 km radius

	Height-for-age z-score				Stunted	
	(1)	(2)	(3)	(4)	(5)	(6)
In utero exposure	−0.022 (0.071)	−0.035 (0.068)	−0.059 (0.071)	−0.070 (0.064)	0.006 (0.024)	0.020 (0.019)
Child controls	No	Yes	Yes	Yes	Yes	Yes
Family controls	No	No	Yes	Yes	Yes	Yes
County time trend	No	No	No	Yes	No	Yes
Observations	7,451	7,451	7,424	7,424	7,424	7,424
R^2	0.335	0.355	0.390	0.396	0.319	0.327

None of the six columns is significant at conventional levels. All regressions include birth cohort and enumeration area fixed effects. Standard errors clustered on a quarter-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent. All regressions weight by the DHS individual sampling weight. Child controls are sex, birth order, first-born status, twin status, and the preceding birth interval. Family controls are mother's education, mother's height, household head age and sex, wealth-quintile indicators, and six-month gestational SPEI. County time trend denotes county-specific linear trends in calendar month of birth. The outcome in columns (5) and (6) is an indicator for a height-for-age z-score below −2, the clinical threshold for stunting.

Table 12: In utero locust exposure and child underweight status, 30 km radius

	Weight-for-age z-score				Underweight	
	(1)	(2)	(3)	(4)	(5)	(6)
In utero exposure	-0.156** (0.063)	-0.166*** (0.061)	-0.167*** (0.064)	-0.190*** (0.062)	-0.002 (0.013)	0.015 (0.014)
Child controls	No	Yes	Yes	Yes	Yes	Yes
Family controls	No	No	Yes	Yes	Yes	Yes
County time trend	No	No	No	Yes	No	Yes
Observations	7,448	7,448	7,421	7,421	7,421	7,421
R^2	0.343	0.361	0.377	0.385	0.275	0.291

The point estimate is negative and significant in columns (1)-(4) but the binary (underweight) columns (5)-(6) are not significant, consistent with dilution similar to Section 5's weight-for-age discussion at the main radius. All regressions include birth cohort and enumeration area fixed effects. Standard errors clustered on a quarter-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent. All regressions weight by the DHS individual sampling weight. Child controls are sex, birth order, first-born status, twin status, and the preceding birth interval. Family controls are mother's education, mother's height, household head age and sex, wealth-quintile indicators, and six-month gestational SPEI. County time trend denotes county-specific linear trends in calendar month of birth. The outcome in columns (5) and (6) is an indicator for a weight-for-age z-score below -2, the clinical threshold for underweight.

Table 13: Treatment by Radius and Health Outcome

Radius	Weight-for-height z-score	Weight-for-age z-score	Height-for-age z-score	Wasted	Treated
10 km	-0.313*** (0.073)	-0.337*** (0.083)	-0.193** (0.092)	0.023** (0.010)	728
15 km	-0.284*** (0.080)	-0.251*** (0.073)	-0.094 (0.089)	0.028*** (0.009)	1,067
20 km	-0.219*** (0.059)	-0.215*** (0.063)	-0.116 (0.071)	0.021** (0.008)	1,333
25 km	-0.186*** (0.058)	-0.194*** (0.061)	-0.111* (0.066)	0.021** (0.008)	1,529
30 km	-0.185*** (0.055)	-0.167*** (0.064)	-0.059 (0.071)	0.019** (0.008)	1,760
35 km	-0.137** (0.057)	-0.125** (0.063)	-0.044 (0.068)	0.016* (0.008)	1,911
40 km	-0.135** (0.056)	-0.134** (0.062)	-0.057 (0.065)	0.015* (0.008)	2,046
45 km	-0.141*** (0.054)	-0.140** (0.061)	-0.060 (0.064)	0.016** (0.008)	2,164
50 km	-0.119** (0.056)	-0.114* (0.064)	-0.046 (0.065)	0.013 (0.008)	2,285

Standard errors clustered on a quarter-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent. Each cell is a separate regression sharing the control set and fixed effects of column (3) of Table 2. Sample sizes are 7,416 for weight-for-height and wasting, 7,421 for weight-for-age, and 7,424 for height-for-age.

Table 14: Falsification tests: purged placebo estimates on weight-for-height, by radius

Radius	Any-form solitary/transiens	Form + stage + NDVI	Out-of-season damaging form
10 km	0.065 (0.103)	—	0.074 (0.068)
15 km	0.086 (0.093)	—	0.022 (0.063)
20 km	0.123 (0.079)	0.284 (0.253)	0.008 (0.078)
25 km	0.139* (0.077)	0.641** (0.301)	0.085 (0.058)
30 km	0.036 (0.098)	0.168 (0.267)	0.085 (0.065)
35 km	0.027 (0.089)	0.005 (0.209)	0.019 (0.063)
40 km	0.088 (0.079)	0.284* (0.167)	−0.045 (0.059)
45 km	0.078 (0.085)	0.201 (0.217)	−0.020 (0.062)
50 km	0.023 (0.072)	0.032 (0.107)	−0.051 (0.061)

Outcome is weight-for-height (WHZ). Standard errors clustered on a quarter-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent. All three columns are purged of overlap with verified (D4) exposure at the same radius, as described in Section 5.3, and share the control set and fixed effects of column (3) of Table 2. Any-form solitary/transiens is solitary and transiens locust records purged of D4 overlap; Form + stage + NDVI additionally requires the crop-stage and vegetation-decline conditions of equation (9) and is not estimated at 10 or 15 km, where purging leaves fewer than 20 children (see Figure 9); Out-of-season damaging form is damaging-form records purged of D4 overlap that arrive outside the susceptible crop stage. 27 nominal cells, 25 estimated after the 20-observation floor; none is negative and significant.

Table 15: Vegetation decline without a locust record: pseudo-event placebo

	Weight-for-height z-score	Height-for-age z-score	Weight-for-age z-score	Died	Share treated
<i>Panel A: 2014 round</i>					
Any decline	0.205 (0.130)	−0.270 (0.173)	0.005 (0.130)	−0.034 (0.023)	0.81
Matched	0.005 (0.045)	0.032 (0.067)	0.017 (0.045)	0.003 (0.009)	0.23
Count	−0.002 (0.015)	0.025 (0.018)	0.011 (0.014)	−0.004 (0.003)	—
<i>Panel B: 2022 round</i>					
Any decline	0.260* (0.137)	−0.038 (0.205)	0.213 (0.183)	0.019 (0.019)	0.78
Matched	−0.161*** (0.059)	−0.003 (0.053)	−0.097* (0.053)	−0.002 (0.008)	0.23
Count	−0.023* (0.013)	−0.038*** (0.014)	−0.036*** (0.014)	0.002 (0.001)	—

Every 16-day NDVI composite in a child’s own cluster is treated as a candidate event and must satisfy the decline condition of equation (8) and the crop-stage condition of equation (5); no locust record is required. *Any decline* is an indicator for at least one qualifying decline during gestation. *Matched* is an indicator for the children whose largest decline during gestation is among the most severe in their round, with the cutoff (−1.59 standard deviations in 2014, −1.76 in 2022) set so that the treated share equals the verified measure’s share at 30 km. *Count* is the number of qualifying declines during gestation. Each cell is a separate regression sharing the control set and fixed effects of column (3) of Table 2. Observations are 7,982 for the anthropometric outcomes and 8,698 for death in 2014, and 7,416, 7,424, 7,421, and 8,057 in 2022. Standard errors clustered on a quarter-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent.

Table 16: Verified exposure and vegetation decline with and without a nearby locust record, weight-for-height, 2022

	(1)	(2)	(3)	(4)
Verified exposure	−0.184*** (0.055)	−0.188*** (0.055)		−0.167*** (0.055)
Matched decline	−0.159*** (0.057)			
Matched decline, locust record within 50 km			−0.217*** (0.066)	−0.195*** (0.065)
Matched decline, no locust record within 50 km		−0.022 (0.075)	−0.034 (0.078)	−0.079 (0.076)
Observations	7,416	7,416	7,416	7,416

Outcome is the weight-for-height z-score. Verified exposure is the main 30 km measure. Matched decline is defined as in Table 15. It is split by whether any locust record of any form fell within 50 km of the child’s cluster during gestation. All columns share the control set and fixed effects of column (3) of Table 2. Standard errors clustered on a quarter-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent.

cluster and birth-cohort structure the main specification uses. Mother’s education, wealth, land ownership, and household size all become indistinguishable from zero. Mother’s height and household head age remain imbalanced after conditioning, both at the 10 percent level. Gestational SPEI and NDVI remain significant after conditioning, in the opposite direction from their raw comparison. Exposed children experience modestly drier and less green gestational windows than their unexposed neighbors within the same cluster and cohort. I address this in Section 5.3.

B Data and Descriptive Statistics

Table 17: Covariate balance, 30 km radius

	Control	Treatment	Raw difference	FE- adjusted	(SE)
<i>Child</i>					
Female	0.490	0.476	−0.014	−0.026	(0.024)
Birth order	3.38	2.75	−0.629***	−0.054	(0.096)
First born	0.242	0.309	0.066***	0.022	(0.023)
Twin	0.031	0.021	−0.010**	—	
Birth interval, prior (months)	46.0	52.8	6.74***	1.53	(2.10)
Birth interval, next (months)	18.5	16.8	−1.70***	—	
<i>Mother and household</i>					
Mother’s education (years)	7.13	9.42	2.30***	−0.045	(0.148)
Mother’s height (cm)	160.8	160.6	−0.184	0.342*	(0.181)
Household head age	38.9	38.3	−0.576*	−1.04*	(0.563)
Female household head	0.291	0.268	−0.023*	−0.020	(0.021)
Household size	6.06	5.38	−0.683***	−0.020	(0.129)
Wealth quintile	2.48	3.00	0.519***	0.021	(0.035)
<i>Assets and access</i>					
Owens land	0.462	0.545	0.082***	−0.014	(0.019)
Owens livestock	0.687	0.662	−0.025*	0.018	(0.017)
Land owned (hectares)	1.07	0.833	−0.238	−0.380	(0.306)
Rural	0.675	0.640	−0.034***	—	
Distance to major road (km)	8.34	3.75	−4.59***	—	
<i>Gestational climate</i>					
SPEI (6 month)	0.997	1.27	0.268***	−0.177**	(0.077)
NDVI z-score	0.678	0.747	0.070***	−0.164***	(0.034)
Observations	5,843	1,760			

Raw difference is the unconditional mean difference, exposed minus unexposed, with significance from a two-sample *t*-test. FE-adjusted is the coefficient on the exposure indicator from regressing each covariate on exposure with the same cluster and birth-cohort fixed effects, sampling weight, and grid-clustered standard error as column (3) of Table 2. ***, **, * denote significance at 1, 5 and 10 percent. Rural and distance to a major road are fixed within a DHS cluster and have no within-cluster variation once the cluster fixed effect is included, so no FE-adjusted coefficient is reported for either. Twin status and the birth interval to the next child likewise return no FE-adjusted coefficient in the same cluster-by-cohort structure, so none is reported for either. Counts in the observation block are for the full sample; the estimation sample for each covariate differs where a covariate has missing values.

Table 18: Wasting Estimates with and without the drought control

Radius	With SPEI-6	Without SPEI-6	Treated
10 km	−0.313*** (0.073)	−0.320*** (0.071)	698
15 km	−0.284*** (0.080)	−0.289*** (0.078)	1,027
20 km	−0.219*** (0.059)	−0.225*** (0.057)	1,288
25 km	−0.186*** (0.058)	−0.192*** (0.057)	1,479
30 km	−0.185*** (0.055)	−0.191*** (0.054)	1,703
35 km	−0.137** (0.057)	−0.144*** (0.055)	1,850
40 km	−0.135** (0.056)	−0.142*** (0.054)	1,978
45 km	−0.141*** (0.054)	−0.149*** (0.053)	2,091
50 km	−0.119** (0.056)	−0.127** (0.055)	2,210

Outcome is the wasting z-score (WHZ). Standard errors clustered on a quarter-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent. *With SPEI-6* repeats the specification of column (3) of Table 2. *Without SPEI-6* is identical except for dropping the six-month gestational SPEI control, the only variable in the specification that measures drought directly. Observations are 7,416 in every cell. Treated counts are the treated children in that estimation sample, which excludes children with missing data and children who are the only child in their cluster.

Table 19: Exposure and Mortality

Radius	Estimate	Treated	<i>p</i>
10 km	0.003 (0.009)	792	0.729
15 km	0.001 (0.008)	1,160	0.878
20 km	0.005 (0.008)	1,449	0.482
25 km	0.002 (0.008)	1,656	0.746
30 km	0.002 (0.007)	1,904	0.739
35 km	−0.001 (0.007)	2,066	0.867
40 km	0.005 (0.007)	2,209	0.515
45 km	0.004 (0.007)	2,338	0.589
50 km	0.003 (0.007)	2,467	0.642

Outcome is an indicator for death before interview. Standard errors clustered on a quarter-degree spatial grid in parentheses. Sample is 8,057 of the 8,213 births in the 2022 round, including children who died or were not measured (Section 5.3). Treated counts are among all 8,213 births. Baseline mortality is 3.2 percent. Each row is a separate regression sharing the control set and fixed effects of column (3) of Table 2.

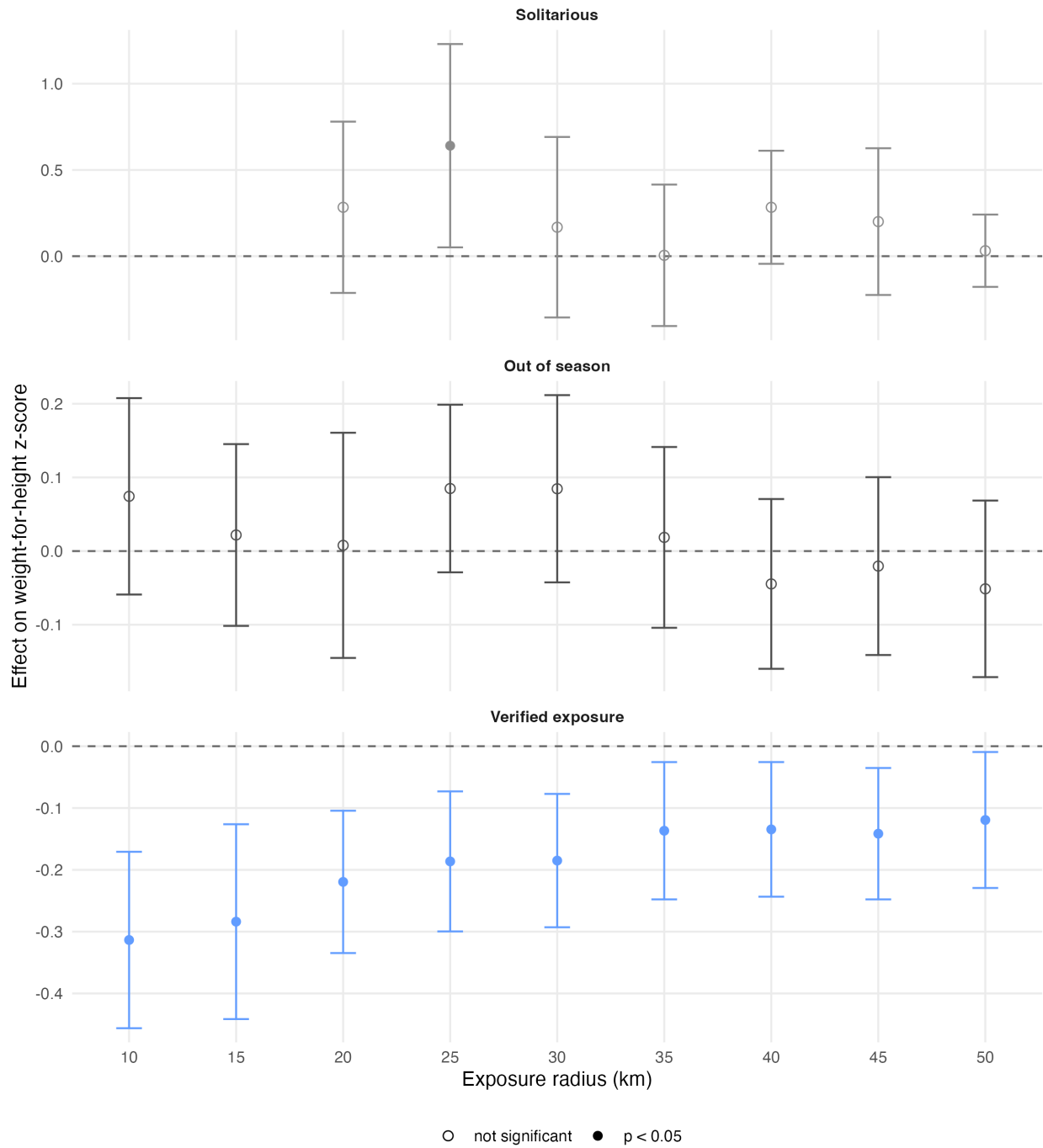


Figure 9: Falsification tests and verified exposure on weight-for-height, by radius. Solitarious and out-of-season estimates are purged of overlap with verified exposure at the same radius, as described in Section 5.3. Filled points are significant at the 5 percent level; hollow points are not. Y-axes are independent across panels and are not comparable to one another in scale. No solitarious estimate is reported at 10 or 15 km: purging leaves 11 and 19 children respectively, both below the 20-observation floor applied throughout this paper, against 56 and 119 before purging (80 and 84 percent of each group is contaminated with verified exposure at that radius).

Table 20: Shifted-calendar 2014 pre-invasion falsification: cells significant at 5 percent

Exposure definition	Height-for-age z-score	Weight-for-age z-score	Weight-for- height z-score	Mortality
Proximity	0/9	1/9	0/9	0/9
+ Damaging form	0/9	0/9	0/9	0/9
+ Susceptible crop stage	0/9	0/9	0/9	0/9
+ Verified vegetation decline	0/9	0/9	0/9	1/9

Each cell counts how many of the nine radii (10 to 50 km) return a coefficient significant at the 5 percent level, out of 144 definition x outcome x radius cells, when the real invasion's calendar pattern is shifted back 93 months onto the 2014 DHS cohort's actual gestation windows (Section 5.3). 2 of 144 cells (1.4 percent) are significant. The one significant mortality cell is negative, at the verified measure and the 30 km radius; the one significant weight-for-age cell is raw proximity at 50 km.

Table 21: In utero and postnatal exposure, weight-for-height, 30 km radius

	(1)	(2)	(3)	(4)
In utero exposure	-0.185*** (0.055)	-0.145** (0.064)	-0.145** (0.064)	
Postnatal exposure, first 12 months		0.106 (0.089)		0.138 (0.102)
Postnatal exposure, months 0 to 6			0.116 (0.086)	
Postnatal exposure, months 6 to 12			0.084 (0.094)	
Sample	All	All	All	Not exposed in utero
Observations	7,416	7,416	7,416	5,513

Outcome is the weight-for-height z-score. Postnatal exposure is a verified damage event within 30 km of the child's cluster after birth, within the stated months and before the interview. All columns share the control set and fixed effects of column (3) of Table 2. Standard errors clustered on a quarter-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent.

Table 22: Exposed children who are underweight versus exposed children who are not, 30 km radius

	Underweight	Not underweight	Difference	<i>p</i>
Mother's education (years)	6.88	9.72	-2.85	< 0.001
Wealth quintile	2.24	3.09	-0.85	< 0.001
Distance to major road (km)	5.15	3.58	1.57	0.001
Rural	0.757	0.626	0.130	< 0.001
Owns land	0.503	0.550	-0.048	0.218
Mother's height (cm)	159.54	160.75	-1.20	< 0.001
Gestational SPEI (6 month)	1.221	1.270	-0.049	0.522
Gestational NDVI z-score	0.877	0.731	0.145	0.015
Age at measurement (months)	15.82	15.66	0.16	0.737

Sample restricted to children flagged as exposed at 30 km. Underweight denotes a weight-for-age z-score below -2 ($N = 189$). Not underweight denotes the remainder of the exposed group ($N = 1,570$). *p*-values from two-sample *t*-tests.

Table 23: Exposed cases versus exposed non-cases, 15 km radius

	Case	Non-case	Difference	<i>p</i>
<i>Panel A: Wasted versus not wasted</i>				
Mother's education (years)	6.20	9.71	−3.51	< 0.001
Wealth quintile	2.29	3.20	−0.91	< 0.001
Distance to major road (km)	5.63	3.42	2.21	0.017
Rural	0.646	0.561	0.085	0.175
Owens land	0.431	0.523	−0.093	0.151
Mother's height (cm)	160.58	160.63	−0.05	0.926
Gestational SPEI (6 month)	1.086	1.216	−0.129	0.342
Gestational NDVI z-score	0.705	0.735	−0.030	0.754
Age at measurement (months)	14.68	15.70	−1.02	0.208
<i>Panel B: Underweight versus not underweight</i>				
Mother's education (years)	6.69	9.86	−3.17	< 0.001
Wealth quintile	2.31	3.26	−0.95	< 0.001
Distance to major road (km)	5.50	3.31	2.19	< 0.001
Rural	0.711	0.549	0.162	< 0.001
Owens land	0.496	0.520	−0.024	0.618
Mother's height (cm)	159.59	160.77	−1.18	0.004
Gestational SPEI (6 month)	1.189	1.210	−0.022	0.826
Gestational NDVI z-score	0.865	0.717	0.148	0.057
Age at measurement (months)	16.10	15.57	0.53	0.393

Sample restricted to children flagged as exposed at 15 km. Panel A compares children with a weight-for-height z-score below −2 ($N = 65$) to the remainder of the exposed group ($N = 1,001$). Panel B compares children with a weight-for-age z-score below −2 ($N = 121$) to the remainder ($N = 946$). *p*-values from two-sample *t*-tests.

Table 24: Heterogeneity: baseline characteristics entered jointly

	Weight-for-height z-score	Height-for-age z-score	Weight-for-age z-score
Agricultural occupation	0.162 (0.117)	0.015 (0.132)	0.112 (0.128)
Owens land	0.030 (0.094)	−0.031 (0.095)	−0.002 (0.097)
Rural	0.206* (0.113)	0.073 (0.122)	0.199* (0.113)
Distance to major road (km)	−0.000 (0.010)	0.016 (0.011)	0.008 (0.010)

Each column is a single regression of the column outcome on exposure at the main radius, all four baseline characteristics, their interactions with exposure, and the control set and fixed effects of column (3) of Table 2; each cell reports one characteristic's exposure interaction from that shared regression, so the four rows in a column are directly comparable. Compare to the one-at-a-time version in the top panel of Table 5: rural residence is the only characteristic that keeps any significance once the other three are controlled for directly. Standard errors clustered on a quarter-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent.

Table 25: Riverine-versus-cropping exposure gap, by radius

Radius (km)	Weight-for-height z-score	Height-for-age z-score	Weight-for-age z-score	Treated in riverine
10	−0.201 (0.131)	−0.211 (0.153)	−0.236 (0.150)	23
15	−0.222 (0.137)	−0.219 (0.172)	−0.264* (0.149)	28
20	−0.268** (0.120)	−0.196 (0.162)	−0.288** (0.145)	28
25	−0.298** (0.121)	−0.204 (0.160)	−0.309** (0.145)	28
30	−0.246* (0.132)	−0.390** (0.191)	−0.369*** (0.135)	35
35	−0.291** (0.135)	−0.406** (0.190)	−0.410*** (0.136)	35
40	0.055 (0.162)	−0.058 (0.220)	−0.001 (0.181)	38
45	0.059 (0.161)	−0.056 (0.219)	0.001 (0.180)	38
50	0.115 (0.180)	−0.047 (0.205)	0.041 (0.164)	39

Each row re-estimates the riverine-versus-cropping interaction from the bottom panel of Table 5 at the row’s radius in place of the main 30 km radius, with the same control set and fixed effects. Standard errors clustered on a quarter-degree spatial grid in parentheses. ***, **, * denote significance at 1, 5 and 10 percent.

B.1 Demographic Health Survey Sample

Table 26 reports summary statistics for the 2022 DHS estimation sample. The anthropometric outcomes carry Kenya’s general burden. Mean height-for-age is -0.91 with a standard deviation of 1.31, and a quarter of the sample sits below -1.77 . Weight-for-height is closer to the reference median at -0.21 , consistent with a population where chronic stunting is more prevalent than acute wasting. Children average 14.7 months at measurement and the sample is 49 percent female. Two thirds of households are rural, 48 percent own land, and 68 percent own livestock, but only 16 percent report an agricultural primary occupation, which is why the occupation variable enters only as a heterogeneity dimension and never as a control.

Distance to a major road is heavily right-skewed. The median cluster sits 2.2 km from one and the 75th percentile 5.8 km, but the maximum exceeds 114 km. Gestational SPEI averages 1.06 at six months, reflecting that the invasion window overlapped a period of above-average moisture in much of the country.

Figure 10 maps the 2022 cluster locations against the major-road network and the World Bank markets used for the price analysis. Clusters follow population density, dense through the western highlands, the central highlands, and the coastal strip, and sparse across the arid north and northeast. The road and market layers follow the same pattern, which is the source of the remoteness gradient that recurs throughout the results.

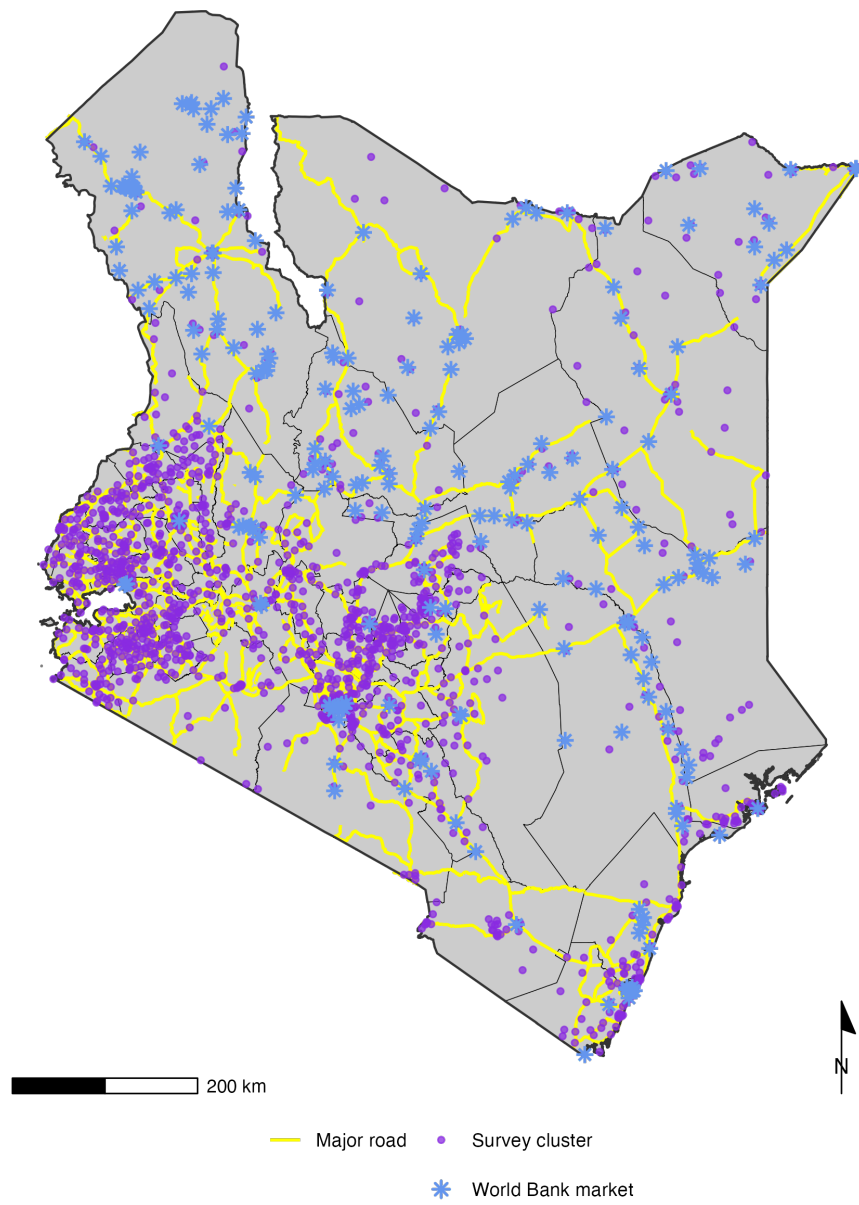


Figure 10: 2022 DHS cluster locations, major roads, and World Bank market locations.

Table 26: Summary statistics, 2022 DHS estimation sample

	Obs.	Mean	St. Dev.	Min	Pctl(25)	Median	Pctl(75)	Max
<i>Anthropometry</i>								
Height-for-age (HAZ)	7,603	-0.91	1.31	-5.90	-1.77	-0.92	-0.11	5.95
Weight-for-age (WAZ)	7,600	-0.64	1.20	-5.29	-1.43	-0.63	0.13	4.85
Weight-for-height (WHZ)	7,595	-0.21	1.20	-5.00	-1.01	-0.20	0.58	4.82
<i>Child and mother</i>								
Age (months)	7,603	14.71	7.26	1.00	9.00	14.00	21.00	30.00
Female	7,603	0.49	0.50	0.00	0.00	0.00	1.00	1.00
First born	7,603	0.26	0.44	0.00	0.00	0.00	1.00	1.00
Twin	7,603	0.03	0.17	0.00	0.00	0.00	0.00	1.00
Birth order	7,603	3.23	2.24	1.00	1.00	3.00	4.00	14.00
Mother's education (yr)	7,603	7.66	5.08	0.00	3.50	8.00	12.00	24.00
Mother's height (cm)	7,603	160.76	4.39	139.70	160.40	160.70	161.00	180.40
<i>Household and location</i>								
Household size	7,603	5.90	2.57	2.00	4.00	5.00	7.00	24.00
Rural	7,603	0.67	0.47	0.00	0.00	1.00	1.00	1.00
Owns land	7,603	0.48	0.50	0.00	0.00	0.00	1.00	1.00
Owns livestock	7,603	0.68	0.47	0.00	0.00	1.00	1.00	1.00
Agricultural occupation	7,603	0.16	0.36	0.00	0.00	0.00	0.00	1.00
Distance to road (km)	7,603	7.28	15.65	0.00	0.74	2.18	5.82	114.65
<i>Gestational window</i>								
SPEI (6 month)	7,598	1.06	1.07	-2.11	0.25	1.20	1.98	3.06
NDVI z-score	7,603	0.69	0.80	-3.04	0.24	0.72	1.19	6.04
Maize price index	7,603	0.95	0.15	0.66	0.83	0.95	1.06	1.58

B.1.1 Sampling Weights

Figure 11 plots the distribution of DHS sampling weights on a log scale. The distribution is right-skewed, with the 99th percentile 6.3 times the median and the maximum 14.2 times, which motivates the winsorized robustness check in Table 27. Winsorizing weights at the first and 99th percentiles and re-running the full grid of 108 regressions leaves the median estimate at -0.074 against -0.090 untrimmed, the negative share at 92 percent against 96 percent, and the significant share at 26 percent against 34 percent. Sample size and composition are identical under winsorizing, since weights are capped rather than observations dropped.

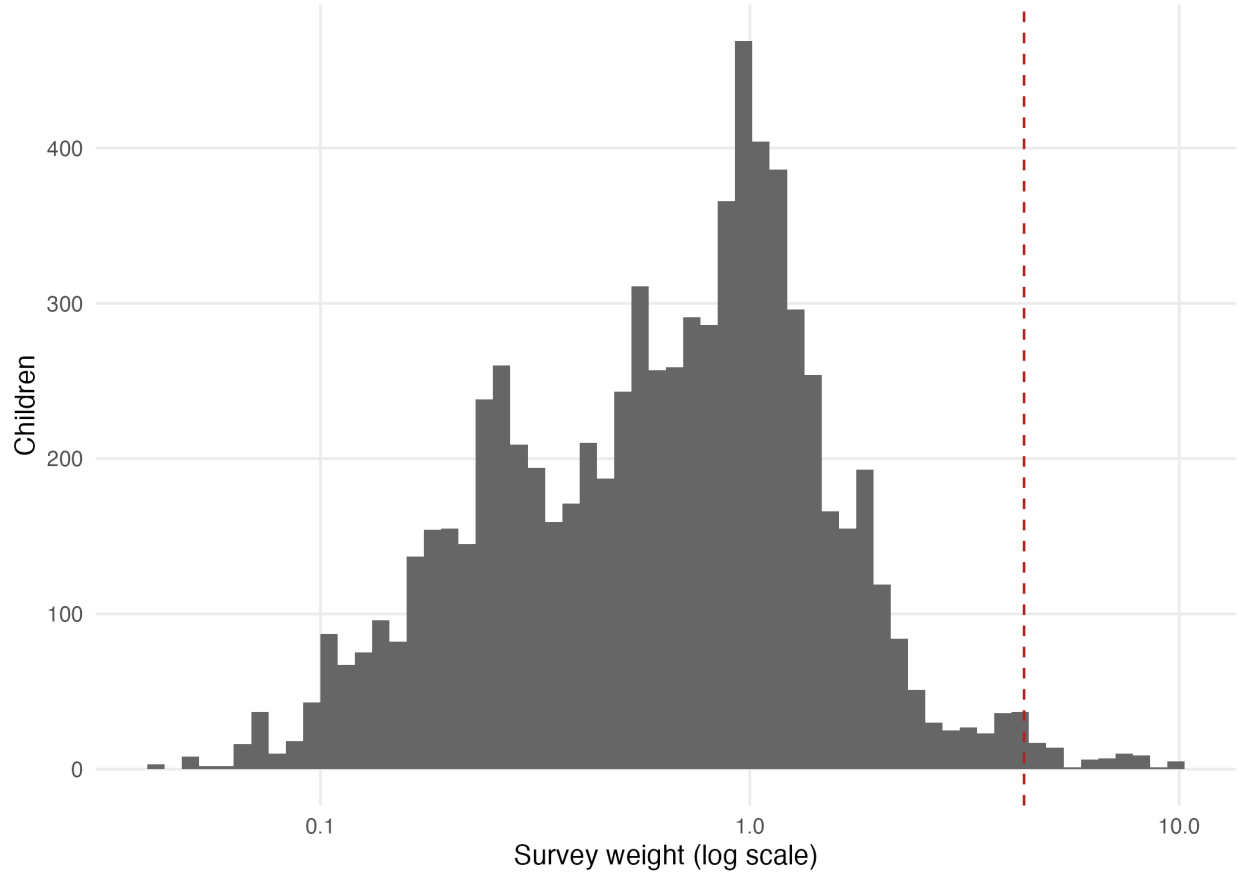


Figure 11: Distribution of DHS sampling weights. Dashed line marks the 99th percentile.

Table 27: Core-grid estimates with untrimmed and winsorized sampling weights

	Untrimmed weights	Winsorized weights
Median estimate	−0.090	−0.074
Mean estimate	−0.095	−0.077
Share of estimates that are negative	0.96	0.92
Share significant at the 5 percent level	0.34	0.26
Regressions	108	108

The grid covers the four exposure definitions (proximity, damaging form, damaging form and crop stage, verified damage), the nine radii from 10 to 50 km, and the three anthropometric z-scores. Each regression shares the control set and fixed effects of column (3) of Table 2. Winsorized weights cap the DHS individual sampling weight at its first and 99th percentiles; the sample is identical in both columns.

B.2 Climate and Vegetation Measures

Rainfall enters through CHIRPS at two temporal resolutions. The monthly product supports the gestational rainfall controls and the SPEI construction. The dekadal product, at ten-day resolution, supports growing-season onset detection. Onset cannot be identified from monthly totals, since a season that begins in the third dekad of March and one that begins in the first dekad of April differ by roughly ten days in the field but

fall in adjacent calendar months, and the crop calendar shift described in Section 3.4 depends on resolving that difference. Extraction is a buffer mean over the DHS displacement radius rather than a point value.

SPEI is constructed at three- and six-month accumulation windows from CHIRPS precipitation and TerraClimate potential evapotranspiration, following [Vicente-Serrano et al. \(2010\)](#). The index standardizes the climatic water balance, precipitation minus PET, against that location’s own historical distribution, so a value of 1.0 means the gestational window was one standard deviation wetter than normal for that cluster rather than wetter than some national average. Only the six-month index enters the control set. The three- and six-month indices are highly correlated within cluster and including both adds collinear variation without adding information, so the longer window is retained as the one more likely to capture a full growing season.

Vegetation enters through MODIS MOD13Q1, a 16-day composite at 250 m resolution. Two features of this product govern how it is used. The compositing window means the date of an NDVI observation is not the date of the underlying satellite pass, since the product selects the best available pixel within each 16-day period. This is why the pre- and post-arrival windows in equation (6) are defined in composites, and why the post-arrival window spans roughly 16 to 48 days depending on where within its composite an arrival falls. Pixels also carry a reliability flag. Pixels flagged as cloud- or snow-contaminated are dropped before averaging to the cluster, so a composite with heavy cloud cover contributes fewer pixels rather than contributing corrupted ones.

NDVI is converted to a per-cluster, per-composite z-score against that cluster’s own pre-invasion climatology for the same composite of the year. This removes both the level difference between a humid highland cluster and an arid northern one and the seasonal cycle within each. A z-score of zero means vegetation was at its normal level for that place at that point in the year. The damage condition in equation (8) then tests a change in this z-score rather than its level, so a cluster that was already dry relative to its own norm does not qualify on dryness alone.

B.3 Crop Calendars

Two sources fix the crop calendar, as Section 3.4 describes. This appendix reproduces an example of each, along with the transcribed calendar those sources are turned into for the crop this paper discusses most, maize.

Figure 12 shows the FEWS NET seasonal calendar for Livelihood Zone 19, the Central Highlands High Potential Zone, which covers the high-rainfall counties north of Nairobi. The zone runs a single extended long-rains season. Land preparation for the maize and bean crop occupies March and April, planting runs from late April into June, and harvesting begins in October and continues to January. Irish potato and cabbage run two cycles. Coffee and tea are harvested across most of the year, and a footnote lists permanent crops including avocado, mango, macadamia, banana, and pineapple. Food prices peak in May and June, in the gap between planting and harvest, and market access is poor during the rains. A swarm arriving in this zone in May or June meets the maize crop at planting, rank 2 in equation (3). A swarm arriving between October and January meets the crop under its harvest bar, rank 3. A swarm arriving in February meets bare ground.

Figure 13 shows the calendar for Livelihood Zone 9, the Northeastern Pastoral Zone, covering Wajir and the surrounding rangeland. The calendar has no crop section. It records camel calving, kidding and lambing, disease outbreaks, milk availability, migration, livestock sales, and the lean season. Households here obtain food from milk, livestock sales, and purchase, and the FEWS assessment calendar tracks livestock condition

rather than crop condition. No date in this zone places a field in a susceptible stage, so equation (5) never holds and no child in this zone is treated. Section 5.4 reports what proximity exposure predicts in these zones, which is nothing.

Figure 14 collects the maize calendar transcribed from calendars like Figure 12, one row per livelihood zone. Zones without a maize row, such as the pastoral zone in Figure 13, are absent from the figure rather than shown blank. The land preparation and planting window moves with each zone's rains, so the bimodal zones on the coast and around Lake Victoria show two separate cropping windows in a single calendar year while the single-season zones, including the Central Highlands zone above, show one.

Seasonal Calendar: Livelihood Zone 19 - Central Highlands, High Potential Zone

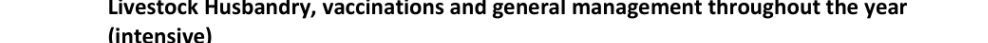
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec					
Seasons	Dry		Long rains									Dry					
Crops																	
Legend	 Land Preparation			 Planting				 Harvesting									
Maize + Beans																	
Coffee																	
Tea																	
Irish potato																	
Cabbage																	
Pyrethrum	 Include pruning																
Others*	 Grown and harvested at different times throughout the year.																
Milk availability	 Throughout the year due to dairy farming																
Livestock	 Livestock Husbandry, vaccinations and general management throughout the year (intensive)																
Food prices					 High												
Market access				 Poor								 Poor					
Labor			 Peak														
Crop Assessments																	
* Permanent crops including avocado, passion fruit, mangos, macadamia nuts, bananas and pineapples																	

Figure 12: FEWS NET seasonal calendar, Livelihood Zone 19, Central Highlands High Potential Zone.

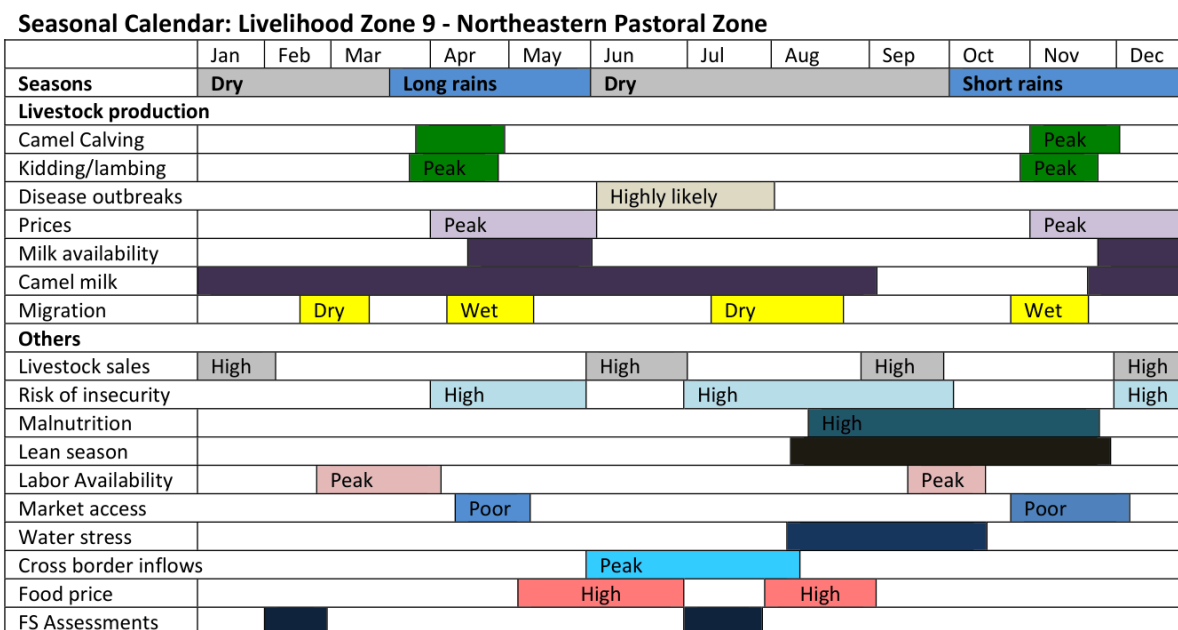


Figure 13: FEWS NET seasonal calendar, Livelihood Zone 9, Northeastern Pastoral Zone. The calendar contains no crop section.

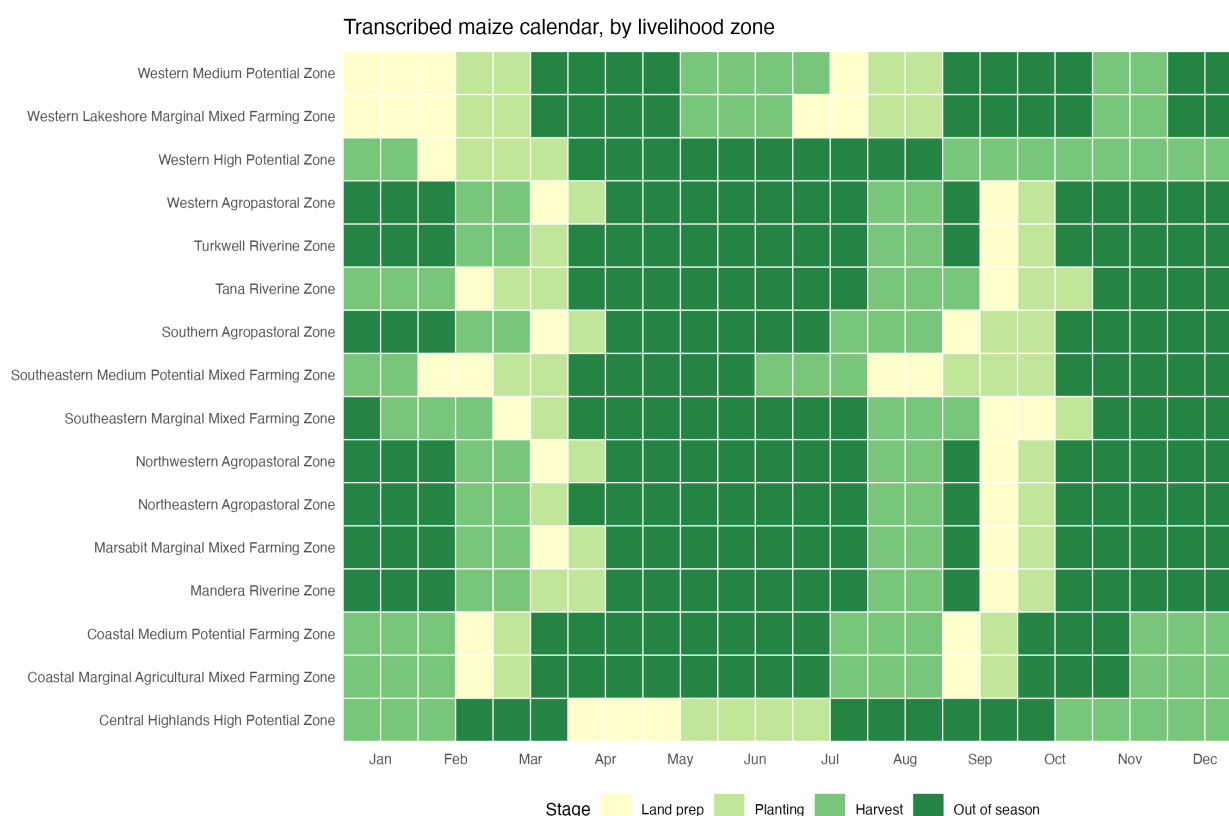


Figure 14: Transcribed maize calendar, by livelihood zone, built from calendars such as Figure 12.

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